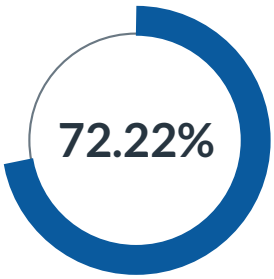
Results

Amanda Kondrat'yev — 1st Attempt



65
Out of 90 points

34:02
Time for this attempt

This assessment has unlimited attempts. Take Now

Attempt History

Results	Points	Score	(Highest score is kept)
Attempt 1	65 of 90	72.22%	(Highest score)

Your Answers:

1 1 / 1 point Multiple Choice

Which of the following tools is used by a technician to remove the jacket or sheathing from a wire when building a cable?

- ○ Cable tester
- ✓ ◉ Cable stripper
- ○ Punchdown tool
- ○ Crimper

Feedback

General Feedback

OBJ 2.8: A cable stripper is used to remove the plastic jacket from an Ethernet cable to access its internal wires (twisted pairs). A cable tester is used to ensure a cable is properly created as a patch cable (straight through) or a crossover cable. Cable testers provide detailed information on the physical and electrical properties of the cable. For example, they test and report cable conditions, crosstalk, attenuation, noise, resistance, and other cable run characteristics. Punchdown tools are used to connect an Ethernet cable to the back of a patch panel, a punchdown block, or the back of a network wall jack. A crimper is a tool used to attach an RJ-45 plastic connector to an unshielded twisted pair (UTP) or shielded twisted pair (STP) cable. It pushes a portion of the plastic into the jacket of the cable to hold it in place.
For support or reporting issues, include Question ID: 63fe04d63b7322449ddbadbc in your ticket. Thank you.

2 0 / 1 point Multiple Choice

Jason has connected a new Roku media player to his high-definition television using an HDMI cable. When he attempts to turn on the Roku device, he sees a message stating "Unauthorized content" on his television. Which of the following should Jason do to fix this issue?

- ○ Turn off the television and Roku, unplug and replug the HDMI cable, and then restart both units
- ✕ ◉ Turn off the Roku and turn it back on
- ○ Turn off the television and turn it back on
- ○ Unplug and replug the HDMI cable from the Roku and television

Feedback

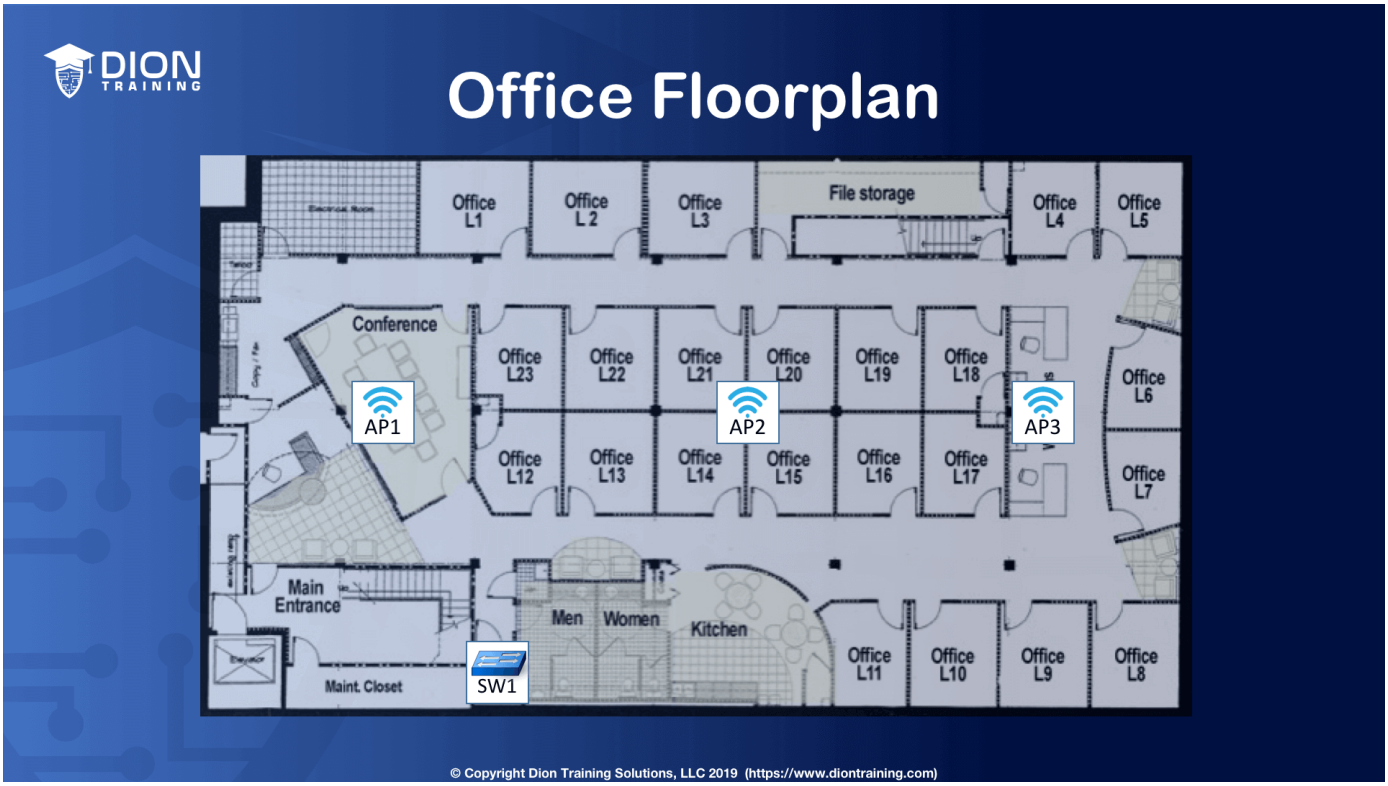
General Feedback

OBJ 5.3: HDMI devices support the use of digital rights management (DRM) using the high-bandwidth digital content protection (HDCP) standard. Sometimes, televisions and displays can fail to properly authenticate themselves with the playback source, such as this Roku device. When this occurs, the best thing to do is shut off both devices, unplug the HDMI cable from both devices, reconnect the HDMI cable to both devices, then turn on the devices one at a time. If this fails to correct the issue, try replacing the HDMI cable with a higher quality cable, as lower quality cables have more issues with allowing the

proper authentication to occur.</p>
For support or reporting issues, include Question ID: 63fe05953b7322449ddbb6e7 in your ticket. Thank you.

3 0 / 1 point Multiple Choice

Wireless network users recently began experiencing speed and performance issues on your network after Access Point 2 (AP2) was replaced due to a recent hardware failure. The original wireless network was installed according to a wireless consultant's specifications and has always worked properly without any past issues.You have been asked to evaluate the situation and resolve any issues you find to improve the network's performance and connectivity. The client has instructed you to adjust the least amount of settings/configurations possible while attempting to fix the issue. Before arriving on-site, you receive the below office's floor plan with an elementary network diagram drawn on top.Based on the information provided to you so far, which network device would you log into first to begin your troubleshooting efforts?



(This is a simulated Performance-Based Question. On the real certification exam, you will be asked to click on the appropriate device in a network diagram to see and modify its configuration.)

- AP1
- AP2

✕ SW1

AP3

Feedback

General Feedback

OBJ 5.5: Since everything was working properly on the network before AP2 was replaced after the recent hardware failure, AP2 likely has some configuration error that has led to the recent connectivity and performance problems. Therefore, you should begin your troubleshooting efforts with AP2. According to the CompTIA troubleshooting method, you should always determine if anything has changed and question the obvious. If AP2 was recently replaced, it is most likely the device with an incorrect configuration setting or the one causing the issues.
For support or reporting issues, include Question ID: 63fe05763b7322449ddbb56b in your ticket. Thank you.

4 1 / 1 point Multiple Choice

IPv4 addresses are written using Base 10 numbers, while IPv6 addresses are written in Base 16 numbers. What type of notation does Base16 utilize?

✓ Hexadecimal

Octet

Decimal

Binary

Feedback

General Feedback

OBJ 2.6: Hexadecimal (or Hex for short) is the system of numbering that uses Base16. This includes the numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F. All IPv6 addresses and MAC addresses are written in hexadecimal format.
For support or reporting issues, include Question ID: 63fe04d83b7322449ddbadd5 in your ticket. Thank you.

5 1 / 1 point Multiple Choice

A developer is using a Platform as a Service (PaaS) environment to build a web application. What is one key advantage of using PaaS for this purpose?

- [x] It provides a pre-configured environment for development, saving time
- [ ] It automates the entire application development process with no input needed
- [ ] It eliminates the need to maintain any part of the application or its code
- [ ] It allows developers to manage physical servers directly for customization

Feedback

General Feedback

OBJ 4.2 - PaaS provides a pre-configured environment for development, including tools, frameworks, and runtime environments, saving time and effort. It does not involve managing physical servers, automating the entire process, or eliminating code maintenance.
For support or reporting issues, include Question ID: 679a45225890cc89a988ce85 in your ticket. Thank you.

6 1 / 1 point Multiple Choice

Which port is used to connect to an SMTP server?

- [x] 25
- [ ] 110
- [ ] 443
- [ ] 143

Feedback

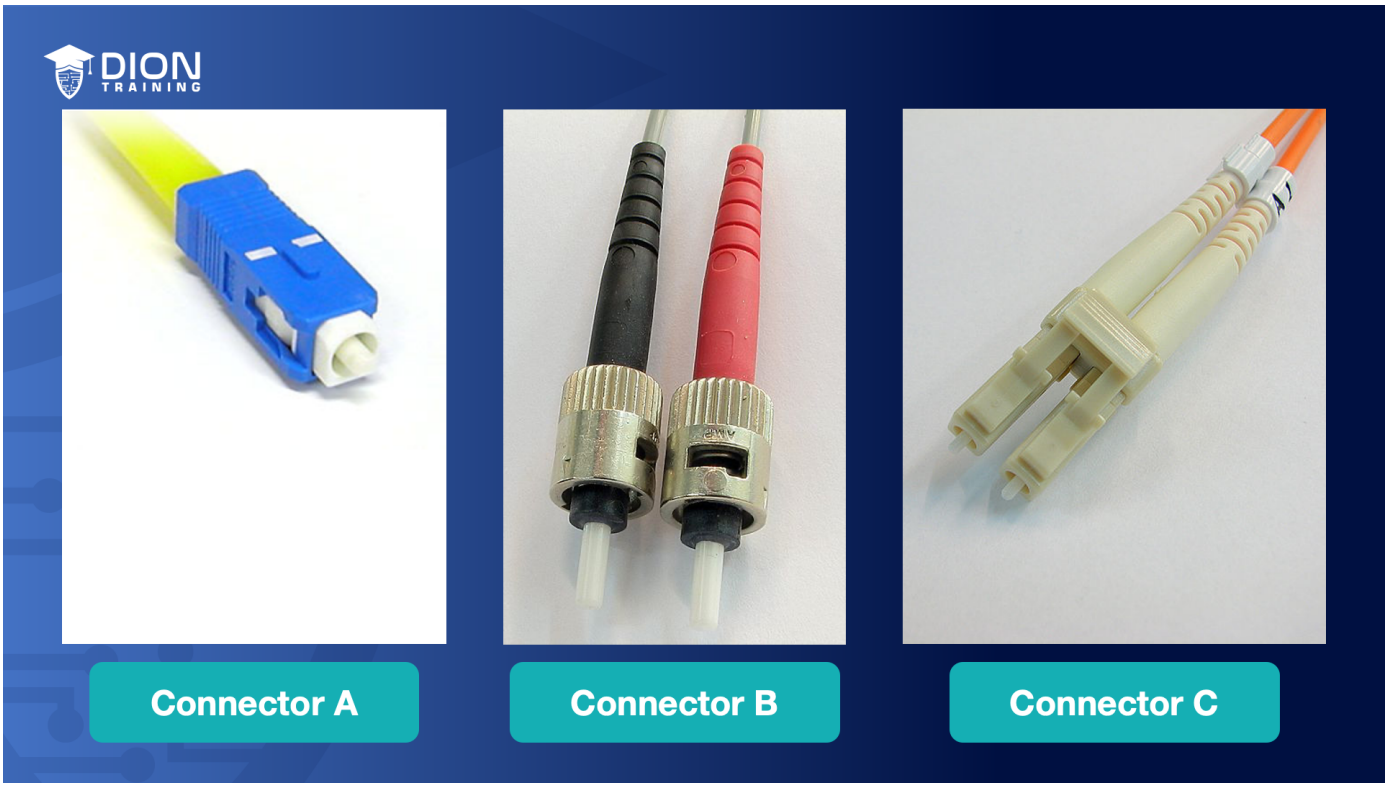
General Feedback

<p>OBJ 2.1: The simple mail transfer protocol (SMTP) is the protocol used to send mail

between hosts on the Internet using TCP port 25. The post office protocol (POP3) is a TCP/IP application protocol providing a means for a client to access email messages stored in a mailbox on a remote server over port 110. The server usually deletes messages once the client has downloaded them. The internet message access protocol (IMAP) is a TCP/IP application protocol that provides a means for a client to access email messages stored in a mailbox on a remote server using TCP port number 143. Unlike POP3, messages persist on the server after the client has downloaded them. IMAP also supports mailbox management functions, such as creating subfolders and access to the same mailbox by more than one client at the same time. The hypertext transfer protocol secure (HTTPS) is a secure protocol used to provide web content to browsers using SSL/TLS encryption over port 443.</p>
For support or reporting issues, include Question ID: 63fe04af3b7322449ddbabdb in your ticket. Thank you.

7 1 / 1 point Multiple Choice

Using the image, which of the following correctly identifies the fiber optic connector shown as Connector A?



(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name on top of the appropriate cable's connector.)

- F
- ST

- [ ] LC
- [x] SC

Feedback

General Feedback

OBJ 3.2 - This is an SC connector. A ST, SC, LC, or MTRJ connector is used to terminate a fiber optic cable. An SC, or subscriber connector, is a general-purpose push/pull style connector that has a square, snap-in connector that latches with a simple push-pull motion. SC connectors are widely used in single-mode fiber optic systems. An ST, or straight tip, connector is a quick-release bayonet-style connector that has a long cylindrical connector. An LC, or Lucent connector, is a small form factor connector that combines the transmit and receive cables into a single square, snap-in connector. The F-type connector is used with coaxial cables, not fiber optic cables.
For support or reporting issues, include Question ID: 63fe05273b7322449ddbb19d in your ticket. Thank you.

8 1 / 1 point Multiple Choice

You just bought a new Virtual Reality headset to use with your computer. To ensure your computer is ready to support the new VR games, which of the following should you install?

- [ ] A larger hard drive
- [ ] A RAID 5 array
- [x] A dedicated graphics card
- [ ] A Bluetooth adapter

Feedback

General Feedback

OBJ 3.5: Virtual Reality (VR) requires a fast processor, lots of memory, and, most importantly, a dedicated graphics card. To support lag-free virtual reality and reduce headaches/eye pain, you need to ensure you have a high-end dedicated graphics card or GPU in your system. Bluetooth is a wireless technology standard used for exchanging data between fixed and mobile devices over short distances using UHF radio waves in the industrial, scientific, and medical radio bands from 2.402 GHz to 2.480 GHz and building

a personal area network (PAN). Bluetooth is commonly used when connecting wireless devices like mice, trackpads, headphones, and other devices. A hard disk drive (HDD) is a device providing persistent mass storage for a workstation or laptop by saving data even when the computer is turned off. Data is stored using platters with a magnetic coating that are spun under disk heads that can read and write to locations on each platter known as sectors.</p>
For support or reporting issues, include Question ID: 63fe05153b7322449ddbb0c1 in your ticket. Thank you.

9 1 / 1 point Multiple Choice

A user brings in their mobile device, reporting that it randomly shuts down even when the battery level shows more than 50%. They also mention that it takes longer than usual to charge the device and that the back of the device feels warmer than normal during charging. Upon inspection, you find no signs of physical damage to the charging port, but the device does not power on without being plugged into a charger. What is the best course of action to resolve the issue?

- [x] Replace the battery after confirming its health and inspecting for swelling or other damage
- [ ] Check the battery settings, enable power-saving mode, and ensure no high-power apps are running in the background
- [ ] Inspect the motherboard and power delivery circuits for potential damage before addressing the battery
- [ ] Test the device with a different charger, reset the device, and verify if the issue persists

Feedback

General Feedback

OBJ 1.1: The best course of action is to replace the battery after confirming its health and inspecting for swelling or other damage. The symptoms (random shutdowns, slow charging, overheating, and reliance on a charger) strongly indicate battery failure. Testing with another charger or resetting the device is unnecessary if the issue persists with a stable power source. Inspecting the motherboard is a later step if replacing the battery does not resolve the issue. Checking battery settings and enabling power-saving mode addresses software inefficiencies but will not fix hardware-related battery failures.
For support or reporting issues, include Question ID: 6797d904aad21b81318d3ef9 in your ticket. Thank you.

10 1 / 1 point Multiple Choice

When using a laser printer, what is used to transfer the image to a piece of paper?

- ◯ Ribbon
- ◯ Ink
- ✓ ◉ Toner
- ◯ Filament

Feedback

General Feedback

OBJ 3.8: Toner is used by a laser printer to transfer the image to a piece of paper. Ink is used by an inkjet printer. Filament is used by a 3-D printer to create its final design. Ribbon is used by an impact printer.
For support or reporting issues, include Question ID: 63fe05263b7322449ddbb18e in your ticket. Thank you.

11 1 / 1 point Multiple Choice

A mobile device is brought to you because it won't turn on. The user mentions that the device worked earlier in the day but now shows no signs of life, even when connected to a charger. What is the most logical order of steps to troubleshoot the issue?

- ◯ Perform a reset by holding both the power and volume down buttons for several seconds
- ✓ ◉ Check the charging port for debris, test with another charger, and inspect the battery for swelling
- ◯ Replace the battery with a new one, then confirm if the device powers on normally
- ◯ Disassemble the device and inspect the motherboard for potential power issues

Feedback

General Feedback

OBJ 1.1: The logical first step is to check the charging port for debris, test with another charger, and inspect the battery for swelling. External factors like a faulty charging cable or clogged port are common issues. Swollen batteries require immediate replacement due to safety concerns. Replacing the battery or inspecting the motherboard should only follow after eliminating simpler causes. Performing a reset can resolve software issues but is unlikely to help if the device shows no signs of power.
For support or reporting issues, include Question ID: 6797d8a0aad21b81318d3ef4 in your ticket. Thank you.

12 1 / 1 point Multiple Choice

Dion Training's file server uses a RAID 5 for its main data store. Recently, the RAID began flashing from green to yellow. Additionally, an alert was received that stated, "RAID ARRAY DISK1 FAILURE." Unfortunately, the company no longer has any spare 1 TB, 5400 RPM Seagate drives, so a technician replaces DISK1 with a 1 TB, 7200 RPM Western Digital drive. When the technician attempts to rebuild the array, the drive stays in degraded operation and continues to alert that "RAID ARRAY DISK1 FAILURE" is occurring. What is the most likely cause of this issue?

- ○ The new drive is not formatted as FAT32
- ✓ ◉ The HDD is incompatible
- ○ Incorrect SCSI ID set
- ○ Bad SATA cable

Feedback

General Feedback

OBJ 5.2: In this scenario, likely, the replacement HDD may not be compatible with the RAID. This will cause the RAID to stay degraded and the alert to continue to display. Whenever you need to replace a disk in a RAID, it is considered a best practice to replace it with the same brand and model of the hard drive. For this reason, most professionals purchase spare hard drives when initially building out their RAID array.
For support or reporting issues, include Question ID: 63fe05633b7322449ddbb480 in your ticket. Thank you.

13 0 / 1 point Multiple Choice

Several users from the Sales & Marketing department have called the service desk

complaining that they cannot access the internet from their workstations. You arrive at the salesperson's workstation, open the command prompt, and enter the command IPCONFIG. The command returns an IPv4 address of 10.1.2.3, a subnet mask of 255.255.255.0, and a default gateway of 10.1.2.1. You attempt to run the command "ping 8.8.8.8" but get the message "Request timed out" displayed four times on the screen. You attempt to ping the IP of the salesperson's workstation sitting next to this workstation and receive four successful replies from the command. You then try to ping the gateway (10.1.2.1) but again receive the message of "Request timed out" on your screen. Which of the following is MOST likely the problem with the salespeople's internet connectivity?

- ○ The router that connects the Sales & Marketing department to the rest of the company is not functioning properly
- ○ The salespeople are getting APIPA addresses from the DHCP server
- ○ The wireless access point in the Sales & Marketing department is not working properly
- ✕ ◉ This is a DNS issue since you cannot access the internet

Feedback

General Feedback

OBJ 5.5: Since you can ping other workstations within the Sales & Marketing department, but you cannot successfully ping the gateway device (the router), this indicates that the router is non-functioning at this time. If the router is not functioning, the workstations will be unable to access any devices or servers outside of the Sales & Marketing department since the router is necessary to connect them to those devices.
For support or reporting issues, include Question ID: 63fe05823b7322449ddbb5fc in your ticket. Thank you.

14 1 / 1 point Multiple Choice

You are configuring a SOHO network that will contain 7 devices, but you only have a single public IP address. Which of the following concepts should be configured to allow the 7 devices to share that single IP when connecting to the internet?

- ○ Perimeter network
- ✓ ◉ NAT
- ○ UPnP

DHCP

Feedback

General Feedback

OBJ 2.6: Network address translation (NAT) is a network service provided by a router or proxy server to map private local addresses to one or more publicly accessible IP addresses. NAT can use static mappings but is commonly implemented as network port address translation (PAT) or NAT overloading, where a few public IP addresses are mapped to multiple LAN hosts using port allocations. The dynamic host control protocol (DHCP) is a protocol used to allocate IP addresses to a host when it joins a network. Universal plug-and-play (UPnP) is a protocol framework allowing network devices to autoconfigure services, such as allowing a game console to request appropriate settings from a firewall. A perimeter network (formerly called a Demilitarized Zone or DMZ) is a portion of a private network connected to the Internet and protected against intrusion. Certain services may need to be made publicly accessible from the Internet (such as a web, email, or Minecraft server), and they should be installed in the perimeter network instead of in your intranet. If communication is required between hosts on either side of a perimeter network, then a host within the perimeter network will act as a proxy to take the request.
For support or reporting issues, include Question ID: 63fe04bf3b7322449ddbaca0 in your ticket. Thank you.

15 0 / 1 point Multiple Choice

A computer technician is configuring a NAS for a company to store their corporate data. The technician has installed four 10 TB hard drives operating at 5400 RPM in a RAID 10 configuration. How much usable storage space is contained in this NAS?

- 40 TB
- × 10 TB
- 20 TB
- 30 TB

Feedback

General Feedback

OBJ 3.4: A RAID 10 combines disk mirroring and disk striping to protect data stored

in the array and requires a minimum of four disks. Since the NAS uses a RAID 10 configuration, it is a stripped and mirrored array. When all four 10 TB HDDs are installed, half of the capacity is used by the mirroring functionality of the RAID 10. Therefore, only 20 TB is available as the usable storage capacity. This configuration will give the array higher speed/performance (due to the stripping) and increase redundancy (due to the mirroring).
For support or reporting issues, include Question ID: 63fe05073b7322449ddbb011 in your ticket. Thank you.

16 1 / 1 point Multiple Choice

You have just finished building a Windows server, installing its operating system, updating its security patches, formatting a dedicated data partition, and creating accounts for all of the company's employees. What is the next thing that must be configured to provide users with the ability to access large multifunction devices across the network?

- File sharing
- Audit logging
- RAID storage
- ✓ Print server

Feedback

General Feedback

OBJ 3.7: Most of the options provided are reasonable next steps, but the question specifically states that you want to allow the users to access multifunction devices on the networks. This indicates that the users will need access to a print server since printers are multifunction devices. A print server is a type of server that connects printers to client computers over a computer network. Print servers act as an intermediary between the computers and printers by accepting print jobs and then sending them to the correct printer when it is ready to accept new jobs. A file server is used to host and control access to shared files and folders for the organization.
For support or reporting issues, include Question ID: 63fe04b23b7322449ddbac07 in your ticket. Thank you.

17 1 / 1 point Multiple Choice

You are troubleshooting a storage issue on a customer's computer. During the computer's

operation, you hear a loud, rhythmic sound. Which of the following storage components is likely failing?

- ○ M.2 PCIe
- ✓ ◉ HDD
- ○ RAM
- ○ SSD

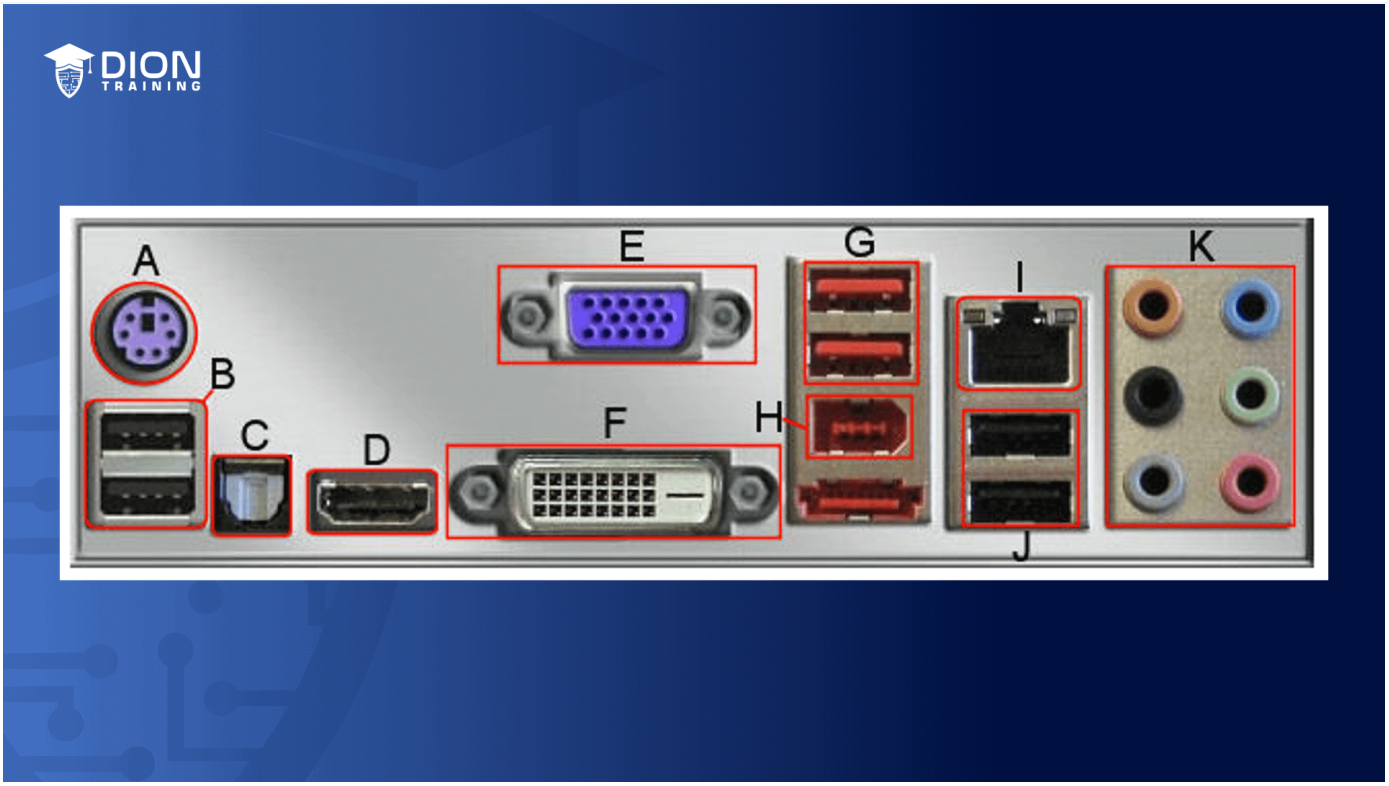
Feedback

General Feedback

OBJ 5.2: A hard disk drive (HDD) is a mechanical storage device that relies on a storage platter and a read arm. When a hard drive begins to fail, the drive will begin to make a loud clicking noise. This noise is usually rhythmic since the failed spot makes the noise every time the drive rotates during the read/write process. The other options (M.2 PCIe, SSD, and RAM) have no moving parts; therefore, no rhythmic noise would be created during their failure.
For support or reporting issues, include Question ID: 63fe059d3b7322449ddbb746 in your ticket. Thank you.

18 0 / 1 point Multiple Choice

Using the image, which of the following correctly indicates the port labeled as J?



(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the port cluster.)

- ◯ USB 2.0
- ✕ ◉ USB 3.0
- ◯ FireWire
- ◯ eSATA

Feedback

General Feedback

OBJ 3.2 - These ports are the USB 2.0 connections. This is indicated by their black color (instead of being red or blue for USB 3.0 ports). Universal serial bus (USB) is the main type of connection interface used on desktops and laptops. A larger Type A connector attaches to a port on the host; Type B and Mini- or Micro-Type B connectors are used for devices. USB 1.1 supports 12 Mbps while USB 2.0 supports 480 Mbps and is backward compatible with 1.1 devices which run at a slower speed. USB devices are considered hot-swappable and each device can draw up to 2.5 W of power over a USB port. USB 3.0 and 3.1 define 5 Gbps (SuperSpeed) and 10 Gbps (SuperSpeed+) rates and can deliver up to 4.5 W of power. For the exam, it is important that you can identify the different parts of a motherboard's rear port cluster by sight.
For support or reporting issues, include Question ID: 63fe05233b7322449ddbb16b in

your ticket. Thank you.

19 1 / 1 point Multiple Choice

A developer is deploying a containerized web application that needs to run consistently across multiple environments, including development, testing, and production. What is one key advantage of using containers for this purpose?

- ◯ Containers allow the application to run without consuming any resources from the host system
- ◯ Containers emulate hardware to simulate the target environment for the application
- ◯ Containers create a full virtual machine to provide isolation for the application
- ✓ ◉ Containers package the application and its dependencies, ensuring it runs the same across all environments

Feedback

General Feedback

OBJ 4.1 - Containers package the application and its dependencies, ensuring consistent behavior across all environments. They do not emulate hardware or create full virtual machines, and they still rely on the host system's resources to operate.
For support or reporting issues, include Question ID: 679a433d2f7beb5108d2ae84 in your ticket. Thank you.

20 1 / 1 point Multiple Choice

Your boss asked you to generate a simple square code that can be easily scanned by a smartphone to access a preconfigured website address or link. Which of the following technologies would BEST meet this requirement?

- ◯ Barcode
- ◯ NFC chip
- ✓ ◉ QR code
- ◯ Smart card

Feedback

General Feedback

OBJ 1.2: In this case, you should select a QR code to meet this requirement. A QR code is a square that contains a specific graphical pattern that can be read by a smartphone's camera. When read, the graphical pattern is converted to data, such as a website address or link that can be easily opened. QR codes can also be read by dedicated QR scanners, as well. A barcode is a pattern of different sized parallel bars, typically representing a product number, such as an ISBN, EAN, or UPC. The reader uses a sensor mechanism (typically either a photodiode, laser, or CCD) to read the intensity of light reflected by the barcode. The reader then reports the number back to application software, which links it to a product database. Near-field communication (NFC) is a set of communication protocols that enable two electronic devices, one of which is usually a portable device such as a smartphone, to establish communication by bringing them within 4 cm of each other. This is commonly used for contactless payment systems, transferring contacts, or transferring a file from one device to another. A smart card, chip card, or integrated circuit card is a physical, electronic authorization device used to control access to a resource. It is typically a plastic credit card-sized card with an embedded integrated circuit chip. In high-security environments, employee badges may contain a smart card embedded chip that must be inserted into a smart card reader to log in or access information on the system.
For support or reporting issues, include Question ID: 63fe049f3b7322449ddbab14 in your ticket. Thank you.

21 0 / 1 point Multiple Choice

Which of the following is an advantage of using a hybrid cloud model?

- ✕ (selected) Unlimited scalability without additional on-premises infrastructure
- Completely managed and insured by a third-party provider
- Lower operational costs compared to public cloud
- Enhanced security by keeping sensitive data on-premises

Feedback

General Feedback

OBJ 4.2: A hybrid cloud model combines both private and public cloud environments, allowing organizations to keep sensitive data and critical workloads on-premises while

leveraging the scalability of public cloud services for less sensitive operations. This model offers better control over security and compliance.
For support or reporting issues, include Question ID: 67973c4722cbe20a3b2640b7 in your ticket. Thank you.

22 0 / 1 point Multiple Choice

You are looking for a device that will proactively stop outside attackers from reaching your internal network. Which of the following devices would be the BEST for you to select?

- IPS
- IDS
- Syslog server
- ✗ Proxy server

Feedback

General Feedback

OBJ 2.5: An intrusion prevention system (IPS) is a form of network security that detects and prevents identified threats. Intrusion prevention systems continuously monitor your network, looking for possible malicious incidents and capturing information about them. An IPS can block malicious network traffic, unlike an IDS, which can only log them. A proxy server is a server application that acts as an intermediary between a client requesting a resource and the server providing that resource. System Logging Protocol (Syslog) uses port 514 and is a way network devices can use a standard message format to communicate with a logging server. It was designed specifically to make it easy to monitor network devices. Devices can use a Syslog agent to send out notification messages under a wide range of specific conditions.
For support or reporting issues, include Question ID: 63fe04b93b7322449ddbac55 in your ticket. Thank you.

23 1 / 1 point Multiple Choice

What RAID level requires at least three hard disks to operate?

- ✓ RAID 5

- RAID 10
- RAID 1
- RAID 0

Feedback

General Feedback

OBJ 3.4: RAID 5 requires at least three hard disks to operate. There is no maximum number of disks required. The parity data is spread across all drives to provide redundancy, which allows the array to operate even if one of the drives fails. RAID 5 is considered one of the most popular and widely used RAID configurations. A RAID 5 provides block-level striping with distributed parity to provide redundancy using a minimum of three disks. A RAID 0 provides disk striping (speed/performance) but not mirroring with a minimum of two disks. A RAID 1 provides mirroring (redundancy) but not disk striping with a minimum of two disks. A RAID 10 combines disk mirroring and disk striping to protect data stored in the array and requires a minimum of four disks.
For support or reporting issues, include Question ID: 63fe051f3b7322449ddbb139 in your ticket. Thank you.

24 1 / 1 point Multiple Choice

A user complains that their USB-C external drive is transferring files to their laptop very slowly, even though the drive and cable support USB 3.2 speeds. After testing the external drive on a different laptop, it achieves normal transfer speeds. What could be the most likely cause of the issue on the original laptop?

- The USB-C cable is faulty and needs to be replaced
- The external drive requires additional power to operate at maximum speed
- The external drive is not compatible with the laptop's operating system
- ✓ The laptop's USB-C port supports only USB 2.0 speeds

Feedback

General Feedback

OBJ 1.2: The most likely cause is that the laptop's USB-C port supports only USB 2.0

speeds. Some laptops have USB-C ports that do not fully support high-speed standards like USB 3.2 or Thunderbolt. This would explain the slower file transfers on the original laptop but normal performance on another system. While a faulty cable or power requirements could also be factors, these would typically cause the issue to persist across devices. Compatibility with the operating system is less likely, as basic USB mass storage usually works across platforms.
For support or reporting issues, include Question ID: 6797da508e2e7a98bb10d75e in your ticket. Thank you.

25 1 / 1 point Multiple Choice

What is the MOST common connector type used when connecting a USB cable to the back of a printer?

- ○ Micro-USB
- ○ Mini-USB
- ○ Type A
- ✓ ◉ Type B

Feedback

General Feedback

OBJ 3.2 - The most common USB connector used to connect to a printer is USB Type B. This connector has a large square-shaped male connector that is connected to the rear of the printer. A USB Type A connector is often used to connect to the workstation or computer itself. This question specifically asked which connector is being attached to the printer. Micro USB is a miniaturized version of the Universal Serial Bus (USB) interface developed for connecting compact and mobile devices such as smartphones, MP3 players, GPS devices, and digital cameras. Mini USB was introduced with USB 2.0 and was used with digital cameras and early tablets.
For support or reporting issues, include Question ID: 63fe050a3b7322449ddbb039 in your ticket. Thank you.

26 0 / 1 point Multiple Choice

Which of the following best describes a Type I hypervisor?

- It is primarily used for desktop virtualization and testing environments
- ✕ It requires a host operating system to function and manage virtualized environments
- It runs on top of an existing operating system to create and manage virtual machines
- It is installed directly on the hardware, offering better performance and security

Feedback

General Feedback

OBJ 4.1: A Type I hypervisor, also known as a "bare-metal" hypervisor, is installed directly on the physical hardware, providing better performance and security by eliminating the need for an underlying operating system. Running on top of an existing operating system describes a Type II hypervisor. Type I hypervisors are typically used in enterprise and data center environments and are not primarily used for desktop virtualization. They also do not rely on an OS to function, so they do not require a host operating system.
For support or reporting issues, include Question ID: 679737cea7a62dd7fe063c9f in your ticket. Thank you.

27 0 / 1 point Multiple Choice

What type of connector provides power to an internal hard drive from the computer's power supply using a 4-pin connector?

- Thunderbolt
- ✕ SATA
- SCSI
- Molex

Feedback

General Feedback

OBJ 3.2 - A Molex connector provides DC power to the various drives inside a computer case. Molex and Mini-Molex are both 4-pins connectors, with Mini-Molex only

being used to support floppy disk drives. The large-sized one is used for hard disk drives, CD-ROM drives, and DVD drives. SATA connectors have 15 pin and 7 pin varieties. Thunderbolt has 20 and 24 pin varieties. SCSI has 50 and 36 pin varieties.
For support or reporting issues, include Question ID: 63fe052f3b7322449ddbb1fc in your ticket. Thank you.

28 0 / 1 point Multiple Choice

A system administrator is tasked with deploying virtualization to remote employees using cloud resources. Which of the following cloud service models would best meet this requirement?

- ✕ IaaS
- PaaS
- SaaS
- DaaS

Feedback

General Feedback

OBJ 4.2: Desktop as a Service (DaaS) provides virtual desktop infrastructure hosted in the cloud, allowing employees to access their work environment remotely from any device. This model simplifies desktop management and enhances security by keeping data centralized in the cloud.
For support or reporting issues, include Question ID: 67973c6c22cbe20a3b2640bc in your ticket. Thank you.

29 1 / 1 point Multiple Choice

A client reports that their workstation is continually crashing when using certain applications on their macOS system. The system will stop working and display a spinning rainbow pinwheel on the screen. Which of the following is the most likely cause of this error?

- ✓ Failure or corruption of the memory
- Failure of the graphics card

- Failure of the processor
- Failure of the power supply

Feedback

General Feedback

OBJ 5.1: On a Mac (OS X) system, the "Pinwheel of Death" is used to indicate an error. The pinwheel appears as an endlessly spinning rainbow mouse pointer displayed on the screen. This usually occurs due to a memory failure or memory corruption in the system. The pinwheel is like the BSOD used in Windows. The Blue Screen of Death (BSOD) is a "stop screen" that appears when the Windows operating system detects a critical error and literally stops the system so that the error doesn't cause any loss of data. All operating systems have some form of "stop screen," but the Windows version got the name BSOD since it was a screen with a full blue background and white text containing troubleshooting information and error codes.
For support or reporting issues, include Question ID: 63fe05943b7322449ddbb6d8 in your ticket. Thank you.

30 1 / 1 point Multiple Choice

You have been asked to install a new wireless network for your company's new branch office. Your boss wants the network to be high-speed and is willing to buy new hardware and network adapters to ensure all the devices can communicate on the new network. Which of the following wireless standards should you install for the fastest data transfer speeds?

- 802.11g
- 802.11a
- 802.11n
- ✓ 802.11ac

Feedback

General Feedback

802.11ac is the fastest of the four standards listed in this question. The 802.11ac (Wireless AC or Wi-Fi 5) standard utilizes a 5 GHz frequency to provide wireless networking at theoretical speeds up to 3.5 Gbps. Wireless AC uses channel bonding to create a single channel of up to 160 MHz to provide additional bandwidth. Wireless AC

uses multi-user multiple-input-multiple-output (MU-MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. The 802.11b (Wireless B) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 11 Mbps. The 802.11g (Wireless G) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 54 Mbps. The 802.11n (Wireless N) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 108 Mbps or a 5.0 GHz frequency to provide wireless networking at speeds up to 600 Mbps. Wireless N supports the use of multiple-input-multiple-output (MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. Wireless N supports channel bonding by combining two 20 MHz channels into a single 40 MHz channel to provide additional bandwidth.
For support or reporting issues, include Question ID: 63fe04e53b7322449ddbae75 in your ticket. Thank you.

31 1 / 1 point Multiple Choice

You have decided that you wanted to install a second operating system on your computer. After installing the OS and rebooting the computer, you see the "Operating System Not Found" error on your display. You verify that the boot.ini file is configured properly, but the error still appears. What is MOST likely causing this error?

- ✓ An incompatible partition is marked as active
- The MBR bootloader was installed accidentally
- An unsupported version of Linux is installed
- Windows Startup services are not properly running

Feedback

General Feedback

OBJ 5.2: This issue may occur if one or more of the following conditions are true: (1) the basic input/output system (BIOS) does not detect the hard disk, (2) the hard disk is damaged, (3) sector 0 of the physical hard disk drive has an incorrect or malformed master boot record (MBR), (4) an incompatible partition is marked as active, or (5) a partition that contains the MBR is no longer active. The only option provided in this list is that an incompatible partition is marked as active.
For support or reporting issues, include Question ID: 63fe059b3b7322449ddbb72d in your ticket. Thank you.

32 1 / 1 point Multiple Choice

A company is implementing biometric authentication for a secure access system. They want a method that analyzes physical traits visible from a distance to avoid the need for close-up interaction. Which biometric technology should they use?

- ◯ Fingerprint recognition
- ◯ Hand geometry scanning
- ◯ Retina scanning
- ✓ ◉ Facial recognition

Feedback

General Feedback

OBJ 1.1 - Facial recognition technology analyzes unique facial features and can often work at a distance, making it suitable for scenarios where close-up interaction may be impractical. Fingerprint recognition and hand geometry scanning require direct physical contact, while retina scanning demands close proximity and alignment with a scanning device, making them less ideal for this purpose.
For support or reporting issues, include Question ID: 6797d086f4ec0def1d268e72 in your ticket. Thank you.

33 1 / 1 point Multiple Choice

You are troubleshooting an older wireless network that is running Wireless G (802.11g). This network appears to have a lot of collisions and interference. You look up the configuration on two of the three access points in the areas and see they are using Channel 1 and Channel 11. To prevent interference and ensure non-overlapping of the channels, which channel should the third access point utilize?

- ✓ ◉ Channel 6
- ◯ Channel 5
- ◯ Channel 7
- ◯ Channel 8

Feedback

General Feedback

OBJ 2.2: With wireless access points that run 2.4 GHz frequencies, you can only select channels between 1 and 11 in the United States. This includes 802.11b, 802.11g, 802.11n, and 802.11ax networks. To prevent overlapping of the channels, you should select channels 1, 6, and 11. By doing so, you can increase the reliability and throughput of your wireless network.
For support or reporting issues, include Question ID: 63fe04c03b7322449ddbacac in your ticket. Thank you.

34 1 / 1 point Multiple Choice

You are troubleshooting a workstation with four identical memory sticks installed, one in each of the slots labeled 1, 2, 3, and 4. The motherboard has a marking near the memory slots that says, "quad-channel support." Based on what you observe, how many channels are being used by the motherboard's memory?

- ◯ Three channels
- ◯ One channel
- ✓ ◉ Four channels
- ◯ Two channels

Feedback

General Feedback

OBJ 3.3: Since the motherboard says it supports quad-channel and four sticks of memory are installed, it would be using four channels. By using a motherboard that supports a quad-channel memory controller, the motherboard is able to transfer up to 256 bits of memory per transaction (4 x 64 bits) to increase the bandwidth available for data transfer.
For support or reporting issues, include Question ID: 63fe05343b7322449ddbb23d in your ticket. Thank you.

35 1 / 1 point Multiple Choice

Your computer keeps losing track of time and the date in the system. Which of the following is MOST likely causing this issue?

- ◯ The memory is corrupted
- ✓ ◉ The CMOS battery is dead and needs to be replaced
- ◯ The power supply is faulty
- ◯ The motherboard's device drivers need to be updated

Feedback

General Feedback

OBJ 5.1: Most motherboards (especially older ones) have a small, coin-sized battery installed on them to power the CMOS RAM when the computer is powered down. This is where the BIOS settings and date/time are saved when power is removed from a workstation. If a computer continually loses track of the day/time, this is a classic sign that the CMOS battery is dead and needs to be replaced. Please note that some newer motherboards have replaced the need for a CMOS battery by replacing it with an area of nonvolatile memory used instead to store the BIOS settings and the date/time.
For support or reporting issues, include Question ID: 63fe056e3b7322449ddbb507 in your ticket. Thank you.

36 1 / 1 point Multiple Choice

You are building a bitcoin mining workstation that will have multiple graphics cards, lots of memory, and an octa-core processor. This computer is expected to generate a lot of heat during its operations. Which of the following solutions would MOST efficiently dissipate all of the heat produced?

- ✓ ◉ Liquid cooling system
- ◯ Passive heat sink
- ◯ High RPM fans
- ◯ Replace the power supply with a more efficient model

Feedback

General Feedback

OBJ 3.5: Since the system described in the question will have multiple graphics cards (GPUs), lots of memory, and an 8-core CPU, it is expected to generate a lot of heat during its crypto mining operation. For any extremely high-powered processing environment that generates a lot of heat (for example, gaming workstations), liquid cooling becomes the MOST efficient way to dissipate the system's heat. A heat sink is a block of copper or aluminum with fins. A cooling effect occurs when convection is achieved using passive cooling as the fins expose a larger surface area to the air around the component. Fans are used to dissipate the heat generated by the processors by increasing the airflow across the components. Replacing the power supply with a more efficient model can reduce some of the heat generated by the power supply, but it will not reduce the heat generated by the CPU or the graphics cards.
For support or reporting issues, include Question ID: 63fe04f53b7322449ddbaf3a in your ticket. Thank you.

37 0 / 1 point Multiple Choice

You are configuring a new printer for a small real estate office. There are only 4 computers in the network, and they are all connected to a single 4-port switch/router/cable modem device. There are no additional open ports on the device and no servers configured within the network. All the computers operate as part of a single workgroup with no domain controller. You need to configure the printer to allow all 4 computers to print to it as long as they are connected to the switch. Which of the following methods would BEST allow the users to print to the printer based on this network's configuration?

- ○ Configure the printer to support cloud printing
- ○ Configure the printer to support Bluetooth printing
- ○ Configure it as a shared printer connected to one of the four workstations
- ✕ ◉ Configure a print server and connect the printer to it

Feedback

General Feedback

OBJ 3.7: Due to the small size of the network and the lack of additional open switch ports, it would be BEST to connect the printer to one of the workstations and configure it as a shared printer. This will allow any of the other workstations to print to the shared printer via the connected workstation. This allows the workstation to act as a print server, which means that the computer must always be left on or the rest of the users would be unable to print. Bluetooth printing is set up as a one-to-one pairing between a single computer and a single printer that is located within 10 feet. Since this is a small

office, they do not have a dedicated server to configure for use as a print server. Cloud printing is only supported by some printers, and the question doesn't specify if this printer supports this feature.
For support or reporting issues, include Question ID: 63fe05293b7322449ddbb1b6 in your ticket. Thank you.

38 1 / 1 point Multiple Choice

You attempt to log on to a Windows workstation, and an error message pops up that says there is a duplicate IP address detected on the network. You restart the workstation, and the same error is displayed when you attempt to log in again. Which of the following is MOST likely causing this error message to occur?

- The workstation was recently moved from another location
- You are logged into another workstation at the same time
- Both wireless and wired connections are enabled on the workstation
- ✓ A static address has been configured on the workstation

Feedback

General Feedback

OBJ 5.5: If you defined a static IP address for a workstation or network device, duplicate IP address conflicts may occur on a DHCP network. To resolve it, convert the workstation or network device with the static IP address to a dynamic assignment or exclude the static IP address from the DHCP scope on the DHCP server. Duplicate IP addresses are not caused by logging into two workstations at once, nor are they caused by moving a workstation.
For support or reporting issues, include Question ID: 63fe056b3b7322449ddbb4e9 in your ticket. Thank you.

39 1 / 1 point Multiple Choice

What is the term for a system that is no longer directly supported by its vendor?

- UTM system
- Embedded system

✓ Legacy system

Patch system

Feedback

General Feedback

OBJ 2.3: A legacy system is no longer directly supported by its vendor. Networks often need to retain hosts running DOS or legacy versions of Windows (8.1 and earlier) or old-style mainframe computers to run services that are too complex or expensive to migrate to a more modern platform. Legacy systems usually work well for what they do (which is why they don't get prioritized for replacement), but they represent very severe risks in terms of security vulnerabilities. Unified threat management (UTM) devices are network appliances that provide all the security functions of a firewall, malware scanner, intrusion detection, vulnerability scanner, data loss prevention, content filtering, and other security devices into a single device or appliance. Patch systems are used to conduct patch management across your organization's network. An embedded system is a computer system that is designed to perform a specific, dedicated function, such as a microcontroller in a medical drip or components in a control system managing a water treatment plant.
For support or reporting issues, include Question ID: 63fe04c03b7322449ddbacb6 in your ticket. Thank you.

40 0 / 1 point Multiple Choice

Dion Training wants to replace any remaining old CRT displays used in the company with a type of screen that produces less voltage and is more energy-efficient. Additionally, the new screen type selected should not contain a backlight. What type of display technology should they select?

✕ LCD

LED

Plasma

OLED

Feedback

General Feedback

OBJ 3.1 - An organic light-emitting diode (OLED) display is a type of LED flat panel display device that uses organic compounds that emit light when subjected to an electric current. An OLED display works without a backlight because it emits visible light. Thus, it can display deep black levels and be thinner and lighter than a liquid crystal display (LCD). In low ambient light conditions (such as a darker room), an OLED screen can achieve a higher contrast ratio than an LCD, regardless of whether the LCD uses cold cathode fluorescent lamps or an LED backlight. An OLED will consume around 40% of an LCD's power, displaying a primarily black image. For most images, it will consume 60–80% of the power of an LCD. A liquid crystal display (LCD) panel is a type of display technology where the image is made up of liquid crystal cells controlled using electrical charges. The LCD panel is backlight by a cold cathode fluorescent (CCFL) bulb. LCD panels are used on laptops and have replaced CRT monitors as the main type of computer display screen for desktops, as well. A light-emitting diode (LED) display is an LCD panel that uses a LED backlight instead of a CCFL backlight to provide its light source. A plasma display is a type of flat panel display that uses small cells containing plasma with ionized gas that responds to electric fields. Plasma displays create excessive heat and are not as energy efficient as an LCD, LED, or OLED display.
For support or reporting issues, include Question ID: 63fe04a03b7322449ddbab28 in your ticket. Thank you.

41 1 / 1 point Multiple Choice

What is a tool that would allow you to easily identify a network cable that was not labeled in the wiring closet during its installation?

- ○ Punchdown tool
- ○ Cable tester
- ✓ ◉ Tone generator and probe
- ○ Multimeter

Feedback

General Feedback

OBJ 2.8: A tone generator and toner probe is an excellent tool that allows you to easily detect which cable goes to which port on a patch panel in a wiring closet. While you could use a cable tester (if both ends of the cable were available to the technician), a tone generator and probe are much more efficient at identifying cables that were not properly labeled. A tone generator is connected to a wall jack and sends a repeating signal over the cable. The probe can then be used to detect which cable is attached to the wall jack by

detecting the signal being sent by the tone generator. The probe needs to be near or touching the cable with the tone generator attached to identify it positively. A cable tester is used to ensure a cable is properly created as a patch cable (straight through) or a crossover cable. Cable testers provide detailed information on the physical and electrical properties of the cable. For example, they test and report cable conditions, crosstalk, attenuation, noise, resistance, and other cable run characteristics. Punchdown tools are used to connect an Ethernet cable to the back of a patch panel, a punchdown block, or the back of a network wall jack. A multimeter can measure the voltage, amperage, and resistance of a circuit. A multimeter can be a hand-held device useful for basic fault finding and field service work or a bench instrument that can measure electricity with a high degree of accuracy.
For support or reporting issues, include Question ID: 63fe04cf3b7322449ddbad6a in your ticket. Thank you.

42 0 / 1 point Multiple Choice

Dion Training just installed a file server in VLAN 101 and their user workstations in VLAN 102. The users are unable to access the new file server from their workstations. Which of the following devices needs to be installed to allow users in VLAN 102 to access the resources in VLAN 101?

- ◯ Router
- ◯ Wireless access point
- ◯ Unmanaged switch
- ✕ ◉ Hub

Feedback

General Feedback

OBJ 5.5: A router or other gateway device must be installed between two VLANs to allow connections to be routed between them. A Virtual LAN (VLAN) is a logically separate network that is created using switching technology. Even though hosts on two VLANs may be physically connected to the same cabling, local traffic is isolated to each VLAN, so they must use a router to communicate. A hub provides no intelligence in its interconnection functions so that whatever is received on one port is repeated out every other port. A wireless access point (WAP), or access point, connects directly to a wired local area network, and the AP then provides wireless connections using Wi-Fi for other devices to use that wired connection. An unmanaged switch operates out-of-the-box without any setup or configuration by switching network traffic based on MAC addresses and putting each switch port into its own collision domain. A hub, access point, and

unmanaged switch are unable to route traffic and therefore would be unable to connect the two VLANs.
For support or reporting issues, include Question ID: 63fe05693b7322449ddbb4cb in your ticket. Thank you.

43 1 / 1 point Multiple Choice

Which of the following ports should a client use to automatically request an IP address from the server?

- 25
- ✓ 67
- 123
- 69

Feedback

General Feedback

OBJ 2.1: The Dynamic Host Configuration Protocol (DHCP) uses port 67 and is a network management protocol used on Internet Protocol (IP) networks for automatically assigning IP addresses and other communication parameters to devices connected to the network using a client-server architecture. Network Time Protocol (NTP) uses port 123 and is a networking protocol for clock synchronization between computer systems over packet-switched, variable-latency data networks. Trivial File Transfer Protocol (TFTP) uses port 69 and is a simple lockstep File Transfer Protocol that allows a client to get a file from or put a file onto a remote host. The Simple Mail Transfer Protocol (SMTP) uses port 25 and is an internet standard communication protocol for electronic mail transmission.
For support or reporting issues, include Question ID: 63fe04c73b7322449ddbad06 in your ticket. Thank you.

44 1 / 1 point Multiple Choice

A developer is using cross-platform virtualization to run a Linux-based development environment on a Windows machine. After updating the virtualization platform, the developer reports that the virtualized environment failed to launch. What should the technician do FIRST to resolve the issue?

- Roll back the virtualization platform to the previous version
- Increase the allocated storage for the virtualized environment
- ✓ Check the virtualization platform's compatibility with the new update
- Reinstall the Linux environment and reconfigure the virtual machine

Feedback

General Feedback

OBJ 4.1 - The first step is to check the virtualization platform's compatibility with the new update. Updates may introduce changes that affect virtualized environments. Reinstalling the Linux environment or rolling back the platform should only be considered if compatibility issues are confirmed. Increasing storage is unrelated unless space issues are explicitly identified.
For support or reporting issues, include Question ID: 679a40212f7beb5108d2ae70 in your ticket. Thank you.

45 1 / 1 point Multiple Choice

An Android user recently cracked their screen and had it replaced. If they are in a dark room, the phone works fine. If the user enters a room with normal lights on, then the phone's display is dim and hard to read. What is MOST likely the problem?

- Defective display
- ✓ Faulty ambient light sensor
- Low battery
- Auto-brightness is enabled

Feedback

General Feedback

<p>OBJ 5.4: The ambient light sensor appears to be broken or malfunctioning. The ambient light sensor may be too sensitive as it is taking in more light than usual. This can occur if the sensor is faulty or if the screen was replaced incorrectly, and the technician forgot to install the black gasket around the ambient light sensor. The auto-brightness

setting being enabled would increase the brightness in a lit room and decrease the brightness in a dark room. If the device has a low battery, it may dim the display to save battery life but it would still be readable. If the display was defective, it would be difficult to read in all light conditions and not just in the bright room.
For support or reporting issues, include Question ID: 63fe05843b7322449ddbb615 in your ticket. Thank you.

46 1 / 1 point Multiple Choice

You are trying to select the best device to install to detect an outside attacker trying to reach into your internal network. The device should log the event, but it should not take any action to stop it. Which of the following devices would be the BEST for you to select?

- IPS
- Proxy server
- ✓ IDS
- Authentication server

Feedback

General Feedback

OBJ 2.5: An intrusion detection system is a device or software application that monitors a network or system for malicious activity or policy violations. Any malicious activity or violation is typically reported to an administrator or collected centrally using a security information and event management system. Unlike an IPS, which can stop malicious activity or policy violations, an IDS can only log these issues and not stop them. An intrusion prevention system (IPS) conducts the same functions as an IDS but can also block or take actions against malicious events. An authentication, authorization, and accounting (AAA) server is a server used to identify (authenticate), approve (authorize), and keep track of (account for) users and their actions. AAA servers can also be classified based on the protocol they use, such as a RADIUS server or TACACS+ server. A proxy server is a server that acts as an intermediary between a client requesting a resource and the server that provides that resource. A proxy server can be used to filter content and websites from reaching a user.
For support or reporting issues, include Question ID: 63fe04e63b7322449ddbae7f in your ticket. Thank you.

47 1 / 1 point Multiple Choice

Dion Training's video editor wants to install a new dedicated GPU using an expansion card. Which of the following should they verify before installing the new card?

- ◯ The power supply's efficiency rating
- ✓ ◉ The power supply's wattage rating
- ◯ The power supply supports a 12 VDC input
- ◯ The number of SATA connectors available

Feedback

General Feedback

OBJ 3.6: Dedicated graphics cards use the most power of any component in a computer. If someone is installing or replacing a graphics card, it is important to compare the current system's power usage to the wattage rating of the power supply to ensure it can support the new graphics card. SATA connectors are not used with graphics cards but are instead used for storage devices and optical drives. The power supply should support 115 VAC or 230 VAC as its input and provide 12 VDC, 5 VDC, and 3.3 VDC as its output. The power supply's efficiency rating would tell the user how much of the input power is being converted to output power, but this is not important in deciding if the power supply can support the new graphics card.
For support or reporting issues, include Question ID: 63fe04fe3b7322449ddbafa7 in your ticket. Thank you.

48 1 / 1 point Multiple Choice

You are troubleshooting a computer that will not boot. You do not hear anything when you attempt to boot the computer, but you smell something that reminds you of burning plastic. Which of the following tools should you use to diagnose the problem with this computer?

- ◯ ESD strap
- ◯ Cable tester
- ✓ ◉ Power supply tester
- ◯ Toner probe

Feedback

General Feedback

OBJ 5.1: The smell of burning plastic and the computer not turning on are symptoms of a power supply issue. Therefore, you should use a power supply (PSU) tester. A burning smell emitting from your laptop indicates that your machine's cooling system is not functioning properly. If your laptop smells like something is burning, this usually means that your computer is overheating. This can be caused by a fan not working properly. For example, if the power supply unit's fan isn't working properly, the unit can overheat, and the wiring in the power supply's transformer can start to burn or melt. This can then cause the power supply unit to fail, too.
For support or reporting issues, include Question ID: 63fe05823b7322449ddbb601 in your ticket. Thank you.

49 1 / 1 point Multiple Choice

Which of the following connection types allows mobile devices to establish short-range wireless communication and is commonly used for data transfer and payment processing?

- ○ Wi-Fi
- ○ Bluetooth
- ○ USB-C
- ✓ ◉ NFC

Feedback

General Feedback

OBJ 1.3: NFC (Near Field Communication) allows mobile devices to communicate over very short distances, typically a few centimeters/inches, making it ideal for contactless payments and quick data transfers. Bluetooth provides wireless communication over longer distances (up to 30 feet) and is commonly used for peripherals like headphones. Wi-Fi supports high-speed wireless connectivity over much longer distances and is primarily used for internet access. USB-C is a wired connection used for charging and data transfer.
For support or reporting issues, include Question ID: 67973920a7a62dd7fe063cb8 in your ticket. Thank you.

50 1 / 1 point Multiple Choice

Jason would like the fastest internet available at his house, and he called his local fiber-optic provider. Which of the following offers would provide the fastest service to Jason's home network?

- HFC
- FTTN
- FTTC
- ✓ FTTH

Feedback

General Feedback

OBJ 2.7: "Fiber to the X" (FTTx) is commonly used to describe where the fiber connection ends between the service and the subscriber. The closer the fiber is to the user's network, the faster the service. FTTH (fiber to the house) provides fiber directly to the user's home network, making it the fastest option. Traditionally, you will find a 1 Gbps connection or higher with FTTH. FTTN (fiber to the node) or FTTC (fiber to the curb/cabinet) provides fiber only to the local area or neighborhood but then uses copper cabling from the node/cabinet/curb to the home network, which slows down the network (generally, 100-200 Mbps). HFC (Hybrid Fiber Coax) is similar to FTTN/FTTC, except that coaxial cable is used from the cabinet to the home to increase the speed (generally 300-500 Mbps).
For support or reporting issues, include Question ID: 63fe04b23b7322449ddbabfc in your ticket. Thank you.

51 1 / 1 point Multiple Choice

What type of cable is most commonly used to terminate a serial connection to a modem?

- ✓ DB-9
- RJ-45
- RJ-11
- LC

Feedback

General Feedback

OBJ 3.2 - A DB-9 connector is used to terminate an RS-232 serial cable. A ST, SC, LC, or MTRJ connector is used to terminate a fiber optic cable. An RJ-11 connector is used to terminate cables used for phone lines. An RJ-45 connector is used to terminate a copper twisted-pair Ethernet cable.
For support or reporting issues, include Question ID: 63fe05173b7322449ddbb0da in your ticket. Thank you.

52 1 / 1 point Multiple Choice

A technician is troubleshooting an issue with a workstation. The workstation can ping 8.8.8.8 successfully, but it is unable to access Facebook.com. Which internal network server is MOST likely causing these issues?

- ◯ SMTP server
- ◯ DHCP server
- ✓ ◉ DNS server
- ◯ FTP server

Feedback

General Feedback

OBJ 2.4: A DNS server is used to perform name lookups and convert them to valid IP addresses. If a workstation can ping an IP, that indicates the network connection is working. If the workstation cannot access a website using its domain name (like diontraining.com or facebook.com), it indicates a problem might exist with the DNS server. A domain name system (DNS) server is a server that hosts the database of domain names and the IP addresses mapped to those names. Each DNS server is authoritative for certain domain names that their organization owns. A mail server is a server that sends or receives emails on behalf of its users. POP3 and IMAP servers are used to receive email, and SMTP is used to send an email. A dynamic host configuration protocol (DHCP) server is a server configured with a range of addresses to lease. Hosts can be allocated an IP address dynamically or be assigned a reserved IP address based on the host's MAC address. The server can also provide other configuration information, such as the location of DNS servers. A file server is used to host and control access to shared files and folders for the organization. A file transfer protocol (FTP) server is a type of file server that is used to host files for users outside your organization.

For support or reporting issues, include Question ID: 63fe04cc3b7322449ddbad47 in your ticket. Thank you.

53 1 / 1 point Multiple Choice

A customer called the service desk and complained that they could not reach the internet on their computer. You ask the customer to open their command prompt, type in ipconfig, and read you the IP address. The customer reads the IP as 169.254.12.45. What is the root cause of the customer's issue based on what you know so far?

- ◯ Their workstation cannot reach the default gateway
- ✓ ◉ Their workstation cannot reach the DHCP server
- ◯ Their workstation cannot reach the web server
- ◯ Their workstation cannot reach the DNS server

Feedback

General Feedback

OBJ 5.5: Since the customer's IP address is 169.254.12.45, it is an APIPA address. Since the workstation has an APIPA address, it means the DHCP server was unreachable. Automatic Private IP Addressing (APIPA) is a feature of Windows-based operating systems that enables a computer to automatically assign itself an IP address when there is no Dynamic Host Configuration Protocol (DHCP) server available to perform that function. APIPA serves as a DHCP server failover mechanism and makes it easier to configure and support small local area networks (LANs). If no DHCP server is currently available, either because the server is temporarily down or because none exists on the network, the computer selects an IP address from a range of addresses (from 169.254.0.0 - 169.254.255.255) reserved for that purpose.
For support or reporting issues, include Question ID: 63fe058c3b7322449ddbb679 in your ticket. Thank you.

54 1 / 1 point Multiple Choice

A user tries to use their smartphone to make a payment at a store using NFC, but the transaction fails. They mention that they recently updated their smartphone's operating system and have not used NFC since the update. The store's payment terminal successfully processes payments for other customers using NFC. What is the most logical first step to troubleshoot the issue?

- Restart the smartphone to refresh its connection
- Reinstall the payment application on the smartphone
- ✓ Check if NFC is enabled in the smartphone's settings
- Test the payment terminal with another NFC-enabled device

Feedback

General Feedback

OBJ 1.2 - The first step is to check if NFC is enabled in the smartphone's settings. OS updates can sometimes disable certain features, such as NFC, or reset settings to default. Since the terminal is working for other customers, the issue is likely with the smartphone, making this the logical place to start. Restarting the smartphone or reinstalling the payment app can be attempted later if NFC is already enabled.
For support or reporting issues, include Question ID: 6797dc1b8e2e7a98bb10d76f in your ticket. Thank you.

55 1 / 1 point Multiple Choice

What is the primary purpose of a punchdown tool in networking?

- ✓ To secure and trim wires into a punchdown block
- To strip the insulation from network cables
- To crimp an RJ45 connector onto the end of a cable
- To test the signal strength of a network cable

Feedback

General Feedback

OBJ 2.8: The primary purpose of a punchdown tool is to secure and trim wires into a punchdown block. It ensures a reliable connection by pressing the wires into the block's metal connectors. Other tasks, such as stripping insulation and crimping connectors, require different tools.
For support or reporting issues, include Question ID: 67986a6db1664769adf99ab5 in

your ticket. Thank you.

56 1 / 1 point Multiple Choice

A client is attempting to power on their computer, but it keeps booting to the incorrect hard drive. Which of the following actions should you perform to attempt to fix this problem?

- ◯ Press F8 repeatedly during the boot process to enter the boot menu
- ✓ ◉ Reconfigure the boot order within the BIOS/UEFI
- ◯ Unplug any external hard disks
- ◯ Reconfigure the boot order within Windows

Feedback

General Feedback

OBJ 5.2: If the system still fails to boot up from the correct drive, then go into the BIOS/UEFI setup program and check the existing boot order settings. If they are in the wrong order, reconfigure them to ensure the usual boot device is listed first in the list and then restart the computer. There is no need to unplug the external hard disks since you can simply change the boot order in the BIOS. The F8 key will allow the user to choose to boot into Windows in regular or safe mode. This worked by default in Windows 7, 8, and 8.1 but needs to be enabled before it will work in Windows 10. The technician cannot configure the boot order in Windows without first booting into Windows from the hard drive, so the technician would still need to reconfigure the boot order in the BIOS/UEFI first.
For support or reporting issues, include Question ID: 63fe057f3b7322449ddbb5de in your ticket. Thank you.

57 1 / 1 point Multiple Choice

Your friend just noticed their smartphone's battery is running low. Unfortunately, your friend isn't very knowledgeable about computers and smartphones, so they don't know what type of charging cable they need. They know their phone is running iOS 15 and state they obtained their phone back in 2020. What kind of cable does your friend need to charge their phone?

- ◯ Micro USB

- ◯ USB-C
- ◯ Mini USB
- ✓ ◉ Lightning

Feedback

General Feedback

OBJ 3.2 - Lightning cables are used by Apple's iPhone, which are the only smartphones that run the iOS software. Since your friend is using iOS, he uses an iPhone (smartphone) or an older iPad (tablet). Lightning is a proprietary computer bus and power connector created by Apple and is used by every iPhone from the 5 to 13 series. Since 2023, newer Apple devices have shifted to the use of USB-C for charging instead of the lightning cable, though. USB-C is the default standard used in USB 3.1 with a 24-pin connector. The connector is reversible and can be inserted in either direction. Micro USB is a miniaturized version of the Universal Serial Bus (USB) interface developed for connecting compact and mobile devices such as smartphones, Mp3 players, GPS devices, photo printers, and digital cameras. Micro USB connectors exist or have existed in three forms: micro A, micro B, and micro USB 3. Mini USB was introduced with USB 2.0 and was used with digital cameras and early tablets. Mini USB is not commonly used in modern devices.
For support or reporting issues, include Question ID: 63fe04a93b7322449ddbab96 in your ticket. Thank you.

58 1 / 1 point Multiple Choice

While troubleshooting a switch at the Dion Training offices, you believe that one of the ports on a switch is defective. To prove your hypothesis, which of the following tools should you use?

- ◯ Crimper
- ✓ ◉ Loopback plug
- ◯ Punchdown tool
- ◯ Cable tester

Feedback

General Feedback

OBJ 2.8: A loopback plug is used to test a port. It involves connecting pin 1 to pin 3 and pin 2 to pin 6. You can do this either by rewiring the jack or twisting the relevant pairs together on a cable stub. Alternatively, you can purchase a prefabricated loopback plug. When you connect a loopback plug to a port, you should see a solid connection LED. You can also use the loopback plug in conjunction with diagnostic software. A crimper is a tool used to attach an RJ-45 plastic connector to an unshielded twisted pair (UTP) or shielded twisted pair (STP) cable. It pushes a portion of the plastic into the jacket of the cable to hold it in place. A cable tester is used to ensure a cable is properly created as a patch cable (straight through) or a crossover cable. Cable testers provide detailed information on the physical and electrical properties of the cable. For example, they test and report cable conditions, crosstalk, attenuation, noise, resistance, and other cable run characteristics. Punchdown tools are used to connect an Ethernet cable to the back of a patch panel, a punchdown block, or the back of a network wall jack.
For support or reporting issues, include Question ID: 63fe04ea3b7322449ddbaeb3 in your ticket. Thank you.

59 1 / 1 point Multiple Choice

Which of the following ports is used by the Secure Shell (SSH) protocol for secure remote access to network devices?

- ✓ 22
- 443
- 25
- 23

Feedback

General Feedback

OBJ 2.1: SSH (Secure Shell) uses TCP port 22 to securely connect to remote devices over an encrypted session. It is commonly used by network administrators to access servers and network equipment securely.
For support or reporting issues, include Question ID: 67973d3fa7a62dd7fe063cc2 in your ticket. Thank you.

60 1 / 1 point Multiple Choice

Which storage solution provides the most flexibility and scalability for a virtualized environment?

- ◯ Direct-attached storage
- ✓ ◉ Network-attached storage
- ◯ Local solid-state drive
- ◯ External USB drive

Feedback

General Feedback

OBJ 4.1: Network-attached storage (NAS) provides a centralized and scalable storage solution that allows multiple virtual machines to access data over the network efficiently. Direct-attached storage (DAS) is directly connected to a single server and lacks flexibility. A local solid-state drive (SSD) offers high speed but lacks network-wide scalability. An external USB drive is not suitable for enterprise-level virtualization due to limited scalability, accessibility, and speed.
For support or reporting issues, include Question ID: 679738e8a7a62dd7fe063cb3 in your ticket. Thank you.

61 0 / 1 point Multiple Choice

What type of cloud service provides instant access to servers and switches, is quickly provisioned, and is available for management over the internet?

- ◯ IaaS
- ◯ SaaS
- ◯ DaaS
- ✗ ◉ PaaS

Feedback

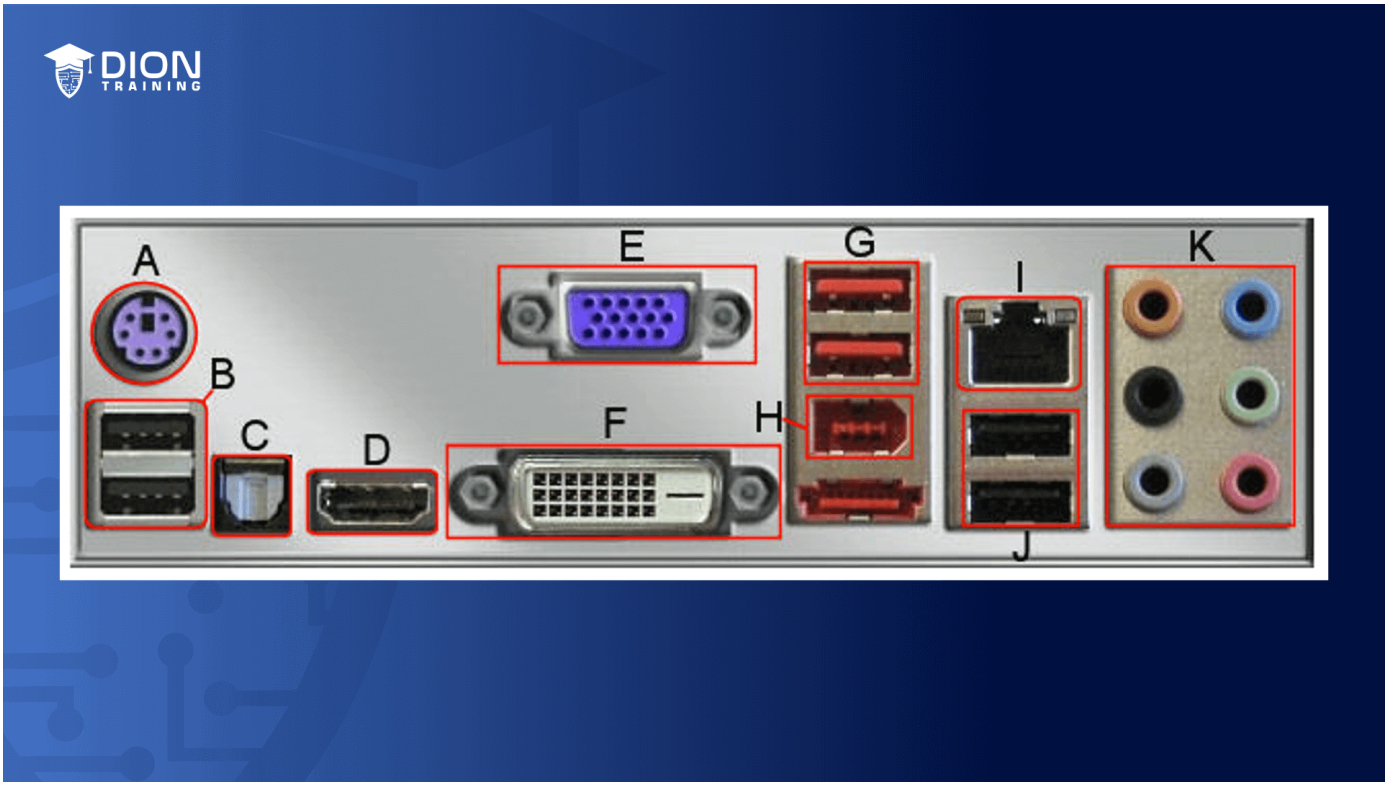
General Feedback

OBJ 4.2: Infrastructure as a service (IaaS) is an instant computing infrastructure

provisioned and managed over the internet. Infrastructure as a Service (IaaS) is a cloud computing service that enables a consumer to outsource computing equipment purchases and running their own data center. IaaS quickly scales up and down with demand, letting you pay only for what you use. It helps you avoid the expense and complexity of buying and managing your physical servers and other data center infrastructure. Each resource is offered as a separate service component, and you only need to rent a particular one for as long as you need it. A cloud computing service provider manages the infrastructure while you purchase, install, configure, and manage your software. Platform as a Service (PaaS) is a cloud computing service that enables consumers to rent fully configured systems that are set up for specific purposes. Software as a Service (SaaS) is a cloud computing service that enables a service provider to make applications available over the Internet to end-users. This can be a calendar, scheduling, invoicing, word processor, database, or other programs. For example, Google Docs and Office 365 are both word processing SaaS solutions. Desktop as a Service (DaaS) provides a full virtualized desktop environment from within a cloud-based service. This is also known as VDI (Virtualized Desktop Infrastructure) and is coming in large enterprise businesses focused on increasing their security and minimizing their operational expenses.</p>
For support or reporting issues, include Question ID: 63fe05453b7322449ddbb30a in your ticket. Thank you.

62 0 / 1 point Multiple Choice

Using the image, which of the following correctly indicates the port labeled as F?



(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the port

cluster.)

- HDMI
- ✕ DVI-I
- DVI-D
- VGA

Feedback

General Feedback

OBJ 3.2 - This port is the DVI-D connection. This is not a DVI-I since there is only a dash and not a cross on the right side of the port. This means it only supports digital and not analog connections. The digital video interface (DVI) came in three models: DVI-A (analog), DVI-D (digital), and DVI-D (analog and digital known as integrated). DVI adapters were designed to replace the VGA ports used in legacy CRT monitors but they were themselves replaced by HDMI. While this is an older technology, you will still see them commonly used in corporate environments. For the exam, it is important that you can identify the different parts of a motherboard's rear port cluster by sight.
For support or reporting issues, include Question ID: 63fe05333b7322449ddbb233 in your ticket. Thank you.

63 0 / 1 point Multiple Choice

A repeat customer is back with their laptop again. This is the third time they have dropped their laptop and caused their hard disk's internal platter to fail. Which of the following devices should you replace their internal hard drive with to prevent this issue from occurring again?

- NVMe
- 5400 RPM HDD
- ✕ Hybrid HDD/SSD
- 7200 RPM HDD

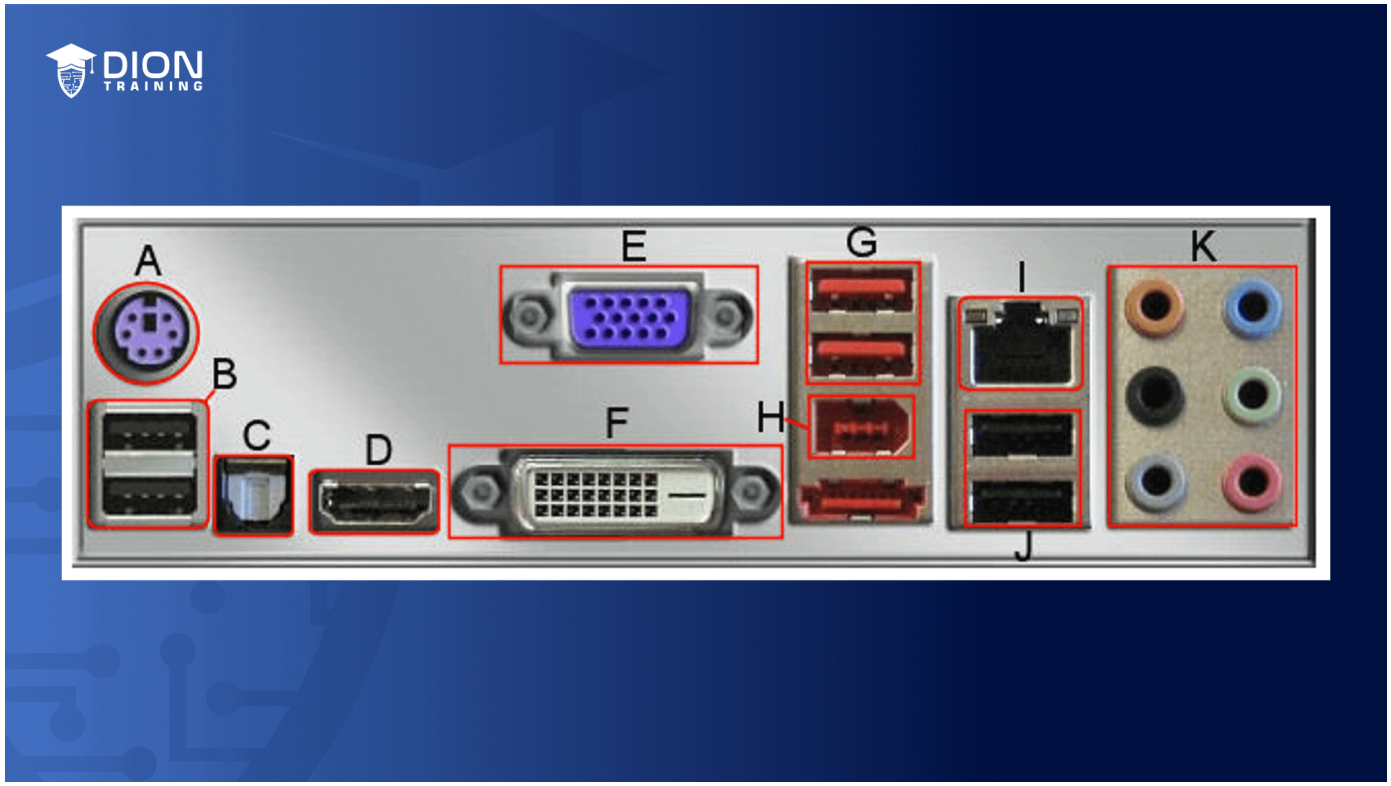
Feedback

General Feedback

OBJ 5.2: Since this customer is accident-prone, you should recommend they install an NVMe or SSD device. These solid-state devices are more resistant to drops since they have no moving parts or platters in them. A hybrid or HDD solution would not be effective, as they both have moving platters in them that can be destroyed by a drop. For support or reporting issues, include Question ID: 63fe05753b7322449ddbb55c in your ticket. Thank you.

64 0 / 1 point Multiple Choice

Using the image, which of the following correctly indicates the port labeled as G?



(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the port cluster.)

- ○ USB-C
- ○ USB 3.0
- ○ FireWire
- ✕ ◉ USB 2.0

Feedback

General Feedback

OBJ 3.2 - These ports are the USB 3.0 connections. The red coloring notates this. USB 3.0 ports are always shown as a different color to differentiate them from the older and slower USB 2.0 ports. Universal serial bus (USB) is the main type of connection interface used on desktops and laptops. A larger Type A connector attaches to a port on the host; Type B and Mini- or Micro-Type B connectors are used for devices. USB 1.1 supports 12 Mbps while USB 2.0 supports 480 Mbps and is backward compatible with 1.1 devices which run at a slower speed. USB devices are considered hot-swappable and each device can draw up to 2.5 W of power over a USB port. USB 3.0 and 3.1 define 5 Gbps (SuperSpeed) and 10 Gbps (SuperSpeed+) rates and can deliver up to 4.5 W of power. For the exam, it is important that you can identify the different parts of a motherboard's rear port cluster by sight.
For support or reporting issues, include Question ID: 63fe05223b7322449ddbb161 in your ticket. Thank you.

65 0 / 1 point Multiple Choice

Which of the following technologies is often used indoors by companies to locate a shopper's position in a store and provide them time-sensitive offers or discounts?

- ✕ ◉ RFID
- ○ GPS
- ○ Cellular
- ○ Bluetooth

Feedback

General Feedback

OBJ 2.2: A mobile device can use its internal GPS, cellular, Bluetooth, or Wi-Fi radios to identify its location. When indoors, the devices can use either Bluetooth or Wi-Fi to determine their location. Bluetooth can be used within a building to provide an accurate location of the device within 3-4 meters. Bluetooth location services are often used to identify where in a store a customer is located and present them with targeted offers. For example, if the store identified that you were in the frozen foods section, they might send you a push notification offering you a discount on a pint of ice cream. Wi-Fi positioning systems use the signal strength or angle of arrival of a signal to determine the device's location within 1-4 meters. The use of Bluetooth and Wi-Fi to determine the device's

location is known as an internal positioning system (IPS). Cellular radios can provide an approximate location based upon the triangulation of your position between different cellular towers. A GPS (global positioning system) device is used to determine a receiver's position on the Earth based on information received from 24 GPS satellites, which operate in a constellation in Medium Earth Orbit (MEO). The receiver must have a line-of-sight to four of the GPS satellites continuously to accurately determine its position on the earth (latitude, longitude, and altitude). The device's GPS receiver can provide an accurate location up to 10 meters.
For support or reporting issues, include Question ID: 63fe04a73b7322449ddbab7d in your ticket. Thank you.

66 1 / 1 point Multiple Choice

You have been asked to run to the store and purchase a network cable for your boss's new laptop. They need a network cable that supports at least 750 Mbps. Which of the following network cables should you purchase to meet the bandwidth required at the lowest price?

- [x] CAT 5e
- [ ] CAT 7
- [ ] CAT 5
- [ ] CAT 6

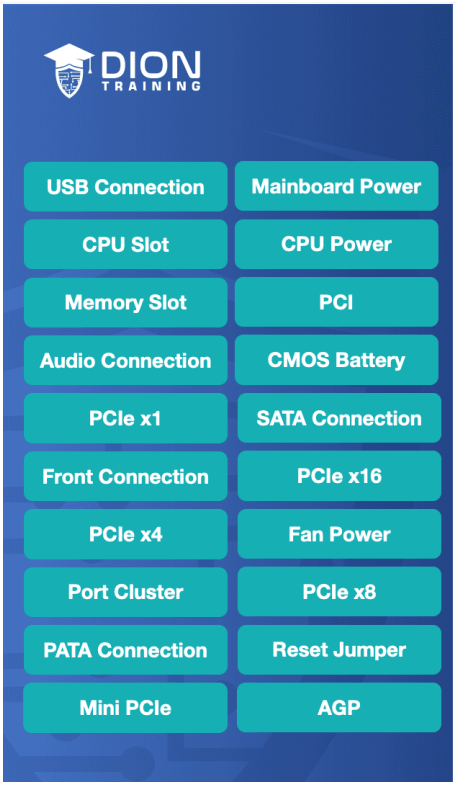
Feedback

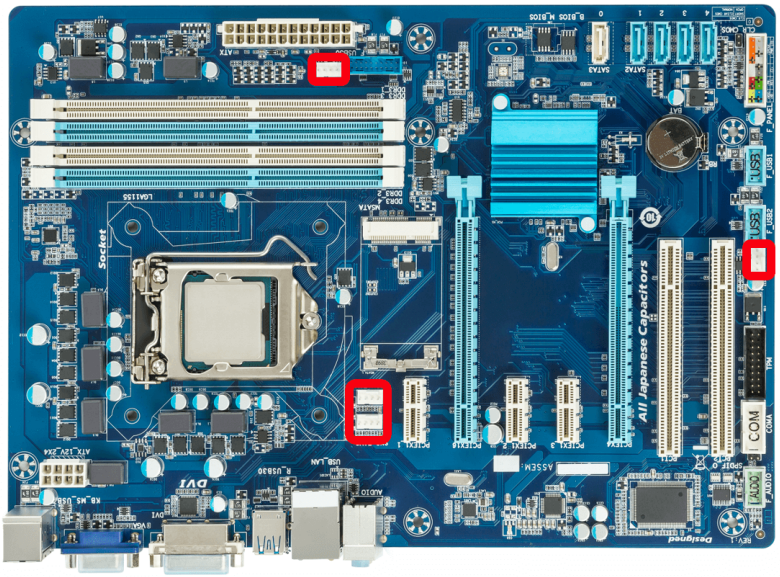
General Feedback

OBJ 3.2 - You should purchase a CAT 5e cable since it can support up to 1000 Mbps (1 Gbps) of bandwidth and is the cheapest option provided that will meet your boss' bandwidth requirements. CAT 5 cables only support 100 Mbps. Therefore, it won't meet the requirements. CAT 6 and CAT 7 cables will support 1 Gbps and 10 Gbps, respectively, but they are more expensive than CAT 5e cables. Therefore, the CAT 5e cable is the best choice for the requirements provided.
For support or reporting issues, include Question ID: 63fe05043b7322449ddbaff3 in your ticket. Thank you.

67 0 / 1 point Multiple Choice

Using the image of the motherboard, which of the following correctly indicates the circled area?





(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the motherboard.)

- AGP
- CPU power
- Fan power
- ✕ Mainboard power

Feedback

General Feedback

OBJ 3.5: The area circled indicates the fan power connectors. The power supply is connected to the motherboard using the mainboard power (24-pin connector at the top of the board), and the case fans connect using these 4-pin connectors to power the fans. For the exam, it is important that you can identify the different parts of the motherboard by sight.
For support or reporting issues, include Question ID: 63fe05333b7322449ddbb22e in your ticket. Thank you.

68 1 / 1 point Multiple Choice

Which of the following devices is used to provide power to a wireless access point?

- ◯ Hub
- ✓ ◉ POE injector
- ◯ Repeater
- ◯ Cable modem

Feedback

General Feedback

OBJ 2.5: A POE (power over ethernet) injector can provide electrical power to a device over a CAT 5e cable. The 802.3af (PoE) standard can support up to 15.4W of power at a distance of up to 100 meters. PoE can support low-powered devices such as VoIP handsets. The 802.3at (PoE+) standard can support up to 25W of power at a distance of up to 100 meters. PoE+ can support higher-powered devices such as PTZ cameras, door controllers, and thin client computers. A hub provides no intelligence in its interconnection functions so that whatever is received on one port is repeated out every other port. This leads to many collisions occurring on the hub and increases the number of rebroadcasts, which slows down the entire network. A repeater is an electronic device that receives a signal and retransmits it. Repeaters are used to extend transmissions so that the signal can cover longer distances or be received on the other side of an obstruction. Repeaters may be wired or wireless, and they rebroadcast whatever signal is received through their transmitter. A cable modem is a type of network bridge that provides bi-directional data communication via radio frequency channels on a hybrid fiber-coaxial (HFC), radio frequency over glass (RFoG), and coaxial cable infrastructure. Cable modems are primarily used to deliver broadband internet access as cable internet, taking advantage of an HFC and RFoG network's high bandwidth.
For support or reporting issues, include Question ID: 63fe04ec3b7322449ddbaec7 in your ticket. Thank you.

69 1 / 1 point Multiple Choice

Due to a global pandemic, your company decides to implement a telework policy for its employees. Unfortunately, the company doesn't have enough time to issue laptops and smartphones to each employee. The Chief Information Officer (CIO) has decided to allow employees to use their laptops and smartphones when conducting their work from home. Which of the following technologies should be implemented to allow the company the ability

to manage the employees' mobile devices, provide security updates, and perform remote administration?

- CDMA
- GSM
- ✓ MDM
- NVMe

Feedback

General Feedback

OBJ 1.3: Mobile application management (MAM) is software that secures and enables IT administrators to have control over enterprise applications on end users' corporate and personal smartphones and tablets. MAM software allows IT administrators to apply and enforce corporate policies on mobile apps and limit the sharing of corporate data among apps. Since the employees will be using their laptops and smartphones, the company should implement and install MDM across the employee's devices to better secure their devices if they give the company permission. The global system for mobile (GSM) communication standard for cellular radio communications and data transfer uses a SIM card to identify the subscriber and network provider. The code division multiple access (CDMA) communication standard for cellular radio communications and data transfer uses a method of multiplexing a communications channel using a code to key the modulation of a particular signal. Non-volatile memory express (NVMe) is an interface for connecting flash memory devices, such as SSDs, directly to a PCI Express bus. NVMe allows much higher transfer rates than SATA/AHCI.
For support or reporting issues, include Question ID: 63fe04a73b7322449ddbab78 in your ticket. Thank you.

70 1 / 1 point Multiple Choice

Which of the following cloud deployment models is dedicated to a single organization and offers enhanced security and control?

- Hybrid cloud
- Community cloud
- Public cloud

Private cloud

Feedback

General Feedback

OBJ 4.2: A private cloud is dedicated to a single organization, offering greater security, control, and compliance compared to public cloud environments. A public cloud provides resources to multiple organizations and individuals, reducing control and security. A community cloud is shared among multiple organizations with common interests or regulatory requirements. Hybrid clouds combine elements of both public and private clouds but do not provide the same level of dedicated control as a private cloud.
For support or reporting issues, include Question ID: 6797350da7a62dd7fe063c84 in your ticket. Thank you.

71 1 / 1 point Multiple Choice

You are troubleshooting a computer that won't display any output on the screen. First, you powered the computer off and on again, noting that the fans were spinning, but there was still no display output. You confirmed that the monitor and video cable were working by testing them with another computer. Next, you reseated the PCIe x16 video card and tried a different video card, but the issue persisted. You checked all power connections inside the case to ensure they were secure. Finally, you tested the computer with minimal components (motherboard, processor, one stick of RAM, and power supply), but there was still no display and no POST beep codes. Given the steps taken, what is the MOST likely cause of the issue?

- The power supply wattage is too low for the components in the computer
- The motherboard has failed and needs to be replaced
- The power supply's voltage switch is set to 240V but the power outlet is providing 110V
- The CMOS battery needs to be replaced

Feedback

General Feedback

OBJ 5.1: If the power supply does not provide enough power, the system may turn on partially (fans spinning) but not fully power the components needed to display output or perform a POST (Power-On Self-Test); therefore, no beep codes would be present. A

motherboard failure is still possible; however, it would most likely cause a POST beep code if the other components are functioning properly. This is less likely because the fans spinning indicate that the motherboard is receiving at least some power as they use low wattage, with the additional findings suggesting a power supply issue. If the power supply's voltage switch is set incorrectly (e.g., to 240V while the power outlet provides 110V), the computer components would likely not turn on at all (no fans spinning), or the power supply would immediately fail. A dead CMOS battery can cause BIOS settings to reset, but it typically does not prevent the computer from displaying output or performing a POST. The computer should still boot with default BIOS settings.
For support or reporting issues, include Question ID: 63fe055b3b7322449ddbb41f in your ticket. Thank you.

72 1 / 1 point Multiple Choice

You have just installed a new hard disk drive into your computer, but the motherboard does not recognize it within the BIOS/UEFI. You have verified the drive is properly connected to the motherboard and the power supply with the correct cables, but it still is not recognized. Which of the following actions would BEST solve this problem?

- [x] Update the firmware of the motherboard
- [ ] Configure a RAID in the BIOS/UEFI
- [ ] Format the new hard drive as FAT-32
- [ ] Format the new hard drive as NTFS

Feedback

General Feedback

OBJ 5.1: This type of error (drive not recognized) is usually the result of the motherboard's firmware not supporting a newer hard drive model or the cables not being properly connected. Since we already established that the cables were properly connected, we have to consider the firmware as the issue. To troubleshoot this issue, you should restart the computer and enter the BIOS/UEFI configuration. If the BIOS/UEFI does not recognize the hard drive, then the motherboard's firmware will need to be updated. If the BIOS/UEFI cannot detect the hard drive, then the operating system cannot detect the drive either (since it relies on the underlying BIOS/UEFI to make the connection). This means that you cannot format the hard drive or configure a RAID.
For support or reporting issues, include Question ID: 63fe055c3b7322449ddbb42b in your ticket. Thank you.

73 1 / 1 point Multiple Choice

A user reports that their laptop's built-in microphone is not picking up sound during video calls. Which of the following should a technician check first?

- ◯ The microphone hardware connection
- ◯ The default playback device settings
- ◯ The system's audio driver status
- ✓ ◉ The application's microphone permissions

Feedback

General Feedback

OBJ 1.1 - Many applications require explicit permission to access a laptop's built-in microphone, and this should be the first item checked. Default playback device settings affect sound output, not input. Audio driver status could cause microphone issues but is less likely to be the immediate cause if the microphone was working previously. Hardware connection issues would not typically apply to a built-in microphone.
For support or reporting issues, include Question ID: 6797d255aad21b81318d3e37 in your ticket. Thank you.

74 0 / 1 point Multiple Choice

Which of the following metrics should you consider to compare the performance of an SSD or HDD?

- ◯ Latency
- ✗ ◉ Seek time
- ◯ IOPS
- ◯ Bandwidth

Feedback

General Feedback

OBJ 5.2: The input/output operations per second (IOPS) is a measurement of performance used to compare a hard disk drive (HDD) and solid-state device (SSD). The IOPS is calculated based on the physical constraints of the media being used. For a hard disk drive, the IOPS is calculated by dividing 1000 milliseconds by the combined average seek time and average latency. The seek time only applies to a traditional hard drive since it calculates the time it takes to move the head to the proper storage location on the platter. For an SSD, there is no seek time used since it can instantly access any portion of the memory without physically moving a read head.
For support or reporting issues, include Question ID: 63fe05783b7322449ddbb57f in your ticket. Thank you.

75 0 / 1 point Multiple Choice

Which of the following cloud services should an organization choose in order to deliver virtual desktops to end-users over the Internet on a per-user license basis?

- ○ IaaS
- ○ SaaS
- ○ DaaS
- × ◉ PaaS

Feedback

General Feedback

OBJ 4.2: Desktop as a Service (DaaS) is a cloud computing offering where a service provider delivers virtual desktops to end-users over the Internet, licensed with a per-user subscription. DaaS is often combined with Virtual Desktop Infrastructure. Platform as a Service (PaaS) is a complete development and deployment environment in the cloud, with resources that enable you to deliver everything from simple cloud-based apps to sophisticated, cloud-enabled enterprise applications. Infrastructure as a Service (IaaS) is a type of cloud computing service that offers essential compute, storage, and networking resources on-demand, on a pay-as-you-go basis. Software as a Service (SaaS) allows users to connect to and use cloud-based apps over the Internet. Common examples are email, calendaring, and office tools (such as Microsoft Office 365). SaaS provides a complete software solution that you purchase on a pay-as-you-go basis from a cloud service provider.
For support or reporting issues, include Question ID: 63fe05413b7322449ddbb2dd in your ticket. Thank you.

76 0 / 1 point Multiple Choice

A client has asked you to build a computer to use as part of their home theater (HTPC) system. The HTPC will be used primarily to stream videos in their living room and must have a small form factor to fit into their entertainment center next to their surround sound system. Which of the following motherboards would be the BEST choice for an HTPC system?

- ITX
- × Micro-ATX
- Mini-ITX
- ATX

Feedback

General Feedback

OBJ 3.5: A form factor means the physical dimensions of a system, and the motherboard form factor defines the size of the case you can use. There is no performance difference between different motherboard form factors, except for the size and number of expansion slots. Mini-ITX is the most commonly used form factor with HTPC systems because they require the smallest fans or passive cooling, which minimizes the noise produced by the system. A micro-ATX motherboard produces more heat and is larger than a mini-ITX motherboard. Micro-ATX motherboards also require a larger (and louder) fan to keep the system cooled, so it is a poor choice for an HTPC. An ATX or ITX motherboard is a bad choice since they are larger and would not fit into a traditional HTPC case.
For support or reporting issues, include Question ID: 63fe05053b7322449ddbb002 in your ticket. Thank you.

77 0 / 1 point Multiple Choice

Which type of internet connection allows for high-speed, bidirectional data communication over a hybrid fiber-coaxial (HFC) connection?

- Fiber
- Satellite

× DSL

Cable

Feedback

General Feedback

OBJ 2.7: A cable modem is a type of network bridge that provides bi-directional data communication via radio frequency channels on a hybrid fiber-coaxial (HFC), radio frequency over glass (RFoG), and coaxial cable infrastructure. Cable modems are primarily used to deliver broadband internet access as cable internet, taking advantage of an HFC and RFoG network's high bandwidth. "Fiber to the X" (FTTx) is commonly used to describe where the fiber connection ends between the service and the subscriber. The closer the fiber is to the user's network, the faster the service. FTTH (fiber to the house) provides fiber directly to the user's home network, making it the fastest option. Traditionally, you will find a 1 Gbps connection or higher with FTTH. FTTN (fiber to the node) or FTTC (fiber to the curb/cabinet) provides fiber only to the local area or neighborhood but then uses copper cabling from the node/cabinet/curb to the home network, which slows down the network (generally, 100-200 Mbps). Satellite systems provide far bigger areas of coverage than can be achieved using other technologies. A Very Small Aperture Terminal (VSAT) microwave antenna is aligned to an orbital satellite that can either relay signals between sites directly or via another satellite. DSL is a technology used to transmit multimedia traffic at high-bit rates over twisted-pair copper wire (over ordinary telephone lines). This allows the telecommunications company to connect a user's home to the local switching center using normal telephone lines, then connect that local switching center (using a DSLAM to multiplex the connections) to the central office over a single high-speed cable (such as a fiber connection).
For support or reporting issues, include Question ID: 63fe04e63b7322449ddbae84 in your ticket. Thank you.

78 1 / 1 point Multiple Choice

Dion Consulting Group is working with a new laptop manufacturer. The manufacturer wants to create the thinnest and lightest laptops in the world. To do this, they need to choose the right technology for their displays. Which technology should Dion Consulting recommend to allow for the thinnest laptop displays?

LCD

IPS

LED

✓ OLED

Feedback

General Feedback

OBJ 3.1 - An OLED display works without a backlight because it emits visible light. Thus, it can display deep black levels and be thinner and lighter than a liquid crystal display (LCD). In low ambient light conditions (such as a darker room), an OLED screen can achieve a higher contrast ratio than an LCD, regardless of whether the LCD uses cold cathode fluorescent lamps or an LED backlight. An in-plane switching (IPS) LCD panel uses technology designed to resolve the quality issues inherent in TN panel technology, including strong viewing angle dependence and low-quality color reproduction. A liquid crystal display (LCD) panel is a type of display technology where the image is made up of liquid crystal cells controlled using electrical charges. The LCD panel is backlight by a cold cathode fluorescent (CCFL) bulb. LCD panels are used on laptops and have replaced CRT monitors as the main type of computer display screen for desktops, as well. A light-emitting diode (LED) display is an LCD panel that uses a LED backlight instead of a CCFL backlight to provide its light source.
For support or reporting issues, include Question ID: 63fe049d3b7322449ddbab00 in your ticket. Thank you.

79 1 / 1 point Multiple Choice

You have just installed a second monitor for a bookkeeper's workstation so they can stretch their spreadsheets across both monitors. This would essentially let them use the two monitors as one combined larger monitor. Which of the following settings should you configure?

- Resolution
- Refresh rate
- ✓ Extended mode
- Color depth

Feedback

General Feedback

OBJ 5.3: The extended mode allows the Windows output to be stretched across two

or more monitors as if they were a single monitor. This can be configured under the Display settings in Windows 10. Refresh rate is the measure of how fast an image can be updated on a monitor or display. If a monitor has a lower refresh rate, then blurring and ghosting can occur. Color depth defines how many unique colors can be displayed by the projected image at once. Most monitors have a default or native resolution. When you first connect a monitor to a Windows workstation, this native resolution is detected, and Windows attempts to configure itself automatically. If this creates an imbalance between the two monitors, a technician can adjust the screen's resolution by changing it in the Display settings area of Windows 10.
For support or reporting issues, include Question ID: 63fe05933b7322449ddbb6c9 in your ticket. Thank you.

80 1 / 1 point Multiple Choice

You have been asked to fix a laptop with a screen image that is being displayed upside down. What is the BEST method to fix this issue?

- ◯ Replace the integrated GPU
- ◯ Replace the LCD panel
- ◯ Update the display drivers
- ✓ ◉ Change the orientation of the screen

Feedback

General Feedback

OBJ 5.3: Windows offers a beneficial function for viewing documents or playing pranks on your friends. With a simple key combination, you can rotate your screen in any direction - flip it upside-down or lay it on the side. To change your laptop screen orientation, hold down the Ctrl and Alt keys on your keyboard, then press the up/down/left/right arrow key. You can also change your screen orientation in the Windows display settings. An integrated GPU is part of the laptop's CPU and cannot be easily replaced. The display drivers and the LCD panel are not the cause of the issue since you can see the screen image properly (although it is being displayed upside down).
For support or reporting issues, include Question ID: 63fe04933b7322449ddbaa7e in your ticket. Thank you.

81 1 / 1 point Multiple Choice

A cybersecurity analyst wants to install a network appliance to conduct packet capturing of the network traffic between the router and the firewall on the network. The device should not be installed in line with the network, so it must receive a copy of all traffic flowing to or from the firewall. Which of the following tools is required to meet these requirements?

- Loopback plug
- ✓ Network tap
- Toner probe
- WiFi analyzer

Feedback

General Feedback

OBJ 2.8: A network tap is used to create a physical connection to the network that sends a copy of every packet received to a monitoring device for capture and analysis. A WiFi analyzer can determine the wireless network's signal strength, the frequencies in use, and any possible radio frequency interference. A tone generator is connected to a wall jack and sends a repeating signal over the cable. The probe can then be used to detect which cable is attached to the wall jack by detecting the signal being sent by the tone generator. The probe needs to be near or touching the cable with the tone generator attached to identify it positively. A cable tester is used to ensure a cable is properly created as a patch cable (straight through) or a crossover cable. Cable testers provide detailed information on the physical and electrical properties of the cable. For example, they test and report cable conditions, crosstalk, attenuation, noise, resistance, and other cable run characteristics.
For support or reporting issues, include Question ID: 63fe04e23b7322449ddbae52 in your ticket. Thank you.

82 1 / 1 point Multiple Choice

A salesperson uses their smartphone as a hotspot while traveling. The first week of their trip, their smartphone could download files at 24 Mbps and stream online videos without any problems. Unfortunately, this week their smartphone is only operating at 256 Kbps when they attempt to download a file. Additionally, they are having difficulty watching online videos due to excessive buffering. Which of the following is MOST likely the problem?

- The smartphone is overheating

- [x] The smartphone's data connection is being throttled
- [ ] The smartphone's hotspot is defective
- [ ] The smartphone is not connected to WiFi

Feedback

General Feedback

OBJ 5.4: Throttling occurs when an internet service provider purposely slows down a user's data transmission. If a device is getting lower speeds without any corresponding device issues, it is likely a result of throttling by the service provider. Most smartphone plans come with a limited amount of full-speed bandwidth, after which the connection is throttled to a slower speed until the next month's plan begins.
For support or reporting issues, include Question ID: 63fe05773b7322449ddbb575 in your ticket. Thank you.

83 1 / 1 point Multiple Choice

You are troubleshooting a network connection issue between the wall jack in a user's office and the switch in the communications closet. When you plug a network cable into the wall jack, there is no connection. You check the switch in the communications closet to determine if the wall jack is properly connected to the switch. Unfortunately, none of the cables in the communication closet are labeled. Which of the following tools should you use to determine which cable in the communications closet is connected to the wall jack in the user's office since none of the cables are properly labeled?

- [ ] Multimeter
- [ ] Loopback plug
- [x] Tone generator and probe
- [ ] Punchdown tool

Feedback

General Feedback

OBJ 5.5: A tone generator is connected to a wall jack and sends a repeating signal over the cable. The probe can then be used to detect which cable is attached to the wall

jack by detecting the signal being sent by the tone generator. The probe needs to be near or touching the cable with the tone generator attached to identify it positively. While a multimeter could be used (in resistance mode) to determine if two ends of a cable are attached to the same cable, the distance between the user's office and the communication closet would prevent a multimeter from being used in this case. A tone generator is connected to a wall jack and sends a repeating signal over the cable. The probe can then be used to detect which cable is attached to the wall jack by detecting the signal being sent by the tone generator. The probe needs to be near or touching the cable with the tone generator attached to identify it positively. Punchdown tools are used to connect an Ethernet cable to the back of a patch panel, a punchdown block, or the back of a network wall jack.
For support or reporting issues, include Question ID: 63fe04bb3b7322449ddbac6e in your ticket. Thank you.

84 0 / 1 point Multiple Choice

You have been assigned a trouble ticket for a workstation that is unable to boot. When you arrive at the workstation, you recognize an error message on the screen that reads, "Error loading operating system, BOOTMGR is missing, Press Ctrl+Alt+Del to restart." Which of the following actions should you take first?

- ◯ Reboot the computer into recovery mode and initiate the disk restoral process
- ◯ Reboot the computer into the Windows 10 repair console and enter "Bootrec /fixmbr" at the command prompt
- ✕ ◉ Reboot the computer into the BIOS to see if the boot order is set to HDD
- ◯ Reboot the computer and allow it to load back into Windows

Feedback

General Feedback

OBJ 5.2: To troubleshoot the failed boot, you must reboot the computer and enter into recovery mode. In Windows 10, you will need to boot from the installation disc, select “Repair your computer,” and then enter the command-line interface (CLI). From the CLI, enter the command "bootrec /fixmbr" to repair the master boot record. Once completed, reboot the computer and allow it to load into Windows.
For support or reporting issues, include Question ID: 63fe05853b7322449ddbb629 in your ticket. Thank you.

85 1 / 1 point Multiple Choice

Which of the following network devices is used to put each connected network host in its own collision domain?

- ○ Access point
- ○ Repeater
- ○ Hub
- ✓ ◉ Unmanaged switch

Feedback

General Feedback

OBJ 2.5: A switch is a network device that receives incoming data into a buffer, then the destination MAC address is compared with an address table so that data is only sent out to the port with the corresponding MAC address. In a switched network, each port is in a separate collision domain. A managed switch is a switch with advanced networking functions and security settings that can be enabled and configured by an administrator. A hub provides no intelligence in its interconnection functions so that whatever is received on one port is repeated out every other port. This leads to many collisions occurring on the hub and increases the number of rebroadcasts, which slows down the entire network. A repeater is an electronic device that receives a signal and retransmits it. Repeaters are used to extend transmissions so that the signal can cover longer distances or be received on the other side of an obstruction. Repeaters may be wired or wireless, and they rebroadcast whatever signal is received through their transmitter. A wireless access point (WAP), or access point, connects directly to a wired local area network, and the AP then provides wireless connections using Wi-Fi for other devices to use that wired connection. An AP supports the connection of multiple wireless devices through their one wired connection.
For support or reporting issues, include Question ID: 63fe04b53b7322449ddbac28 in your ticket. Thank you.

86 1 / 1 point Multiple Choice

What type of connector is most often used to connect to an older smartphone?

- ✓ ◉ Micro USB

- USB-C
- Mini-USB
- USB Type B

Feedback

General Feedback

OBJ 3.2 - Micro USB is a miniaturized version of the Universal Serial Bus (USB) interface developed for connecting compact and mobile devices such as smartphones, MP3 players, GPS devices, photo printers, and digital cameras. Micro USB connectors exist or have existed in three forms: micro A, micro B, and micro USB 3. Mini USB was introduced with USB 2.0 and was used with digital cameras and early tablets. Mini USB is not commonly used in modern devices. Recently, though, the industry moved to the USB Type C (USB-C) connector for smartphones and many other devices as it is faster and easier to connect. USB-C is the default standard used in USB 3.1 with a 24-pin connector. The connector is reversible and can be inserted in either direction. USB Type B is a large form factor USB connector usually reserved for connecting to older printers and scanners.
For support or reporting issues, include Question ID: 63fe04f93b7322449ddbaf6e in your ticket. Thank you.

87 1 / 1 point Multiple Choice

There is a crinkling sound coming from the laser printer in the Dion Training offices. During their investigation, a technician observes that the images are printing at an angle on the pages. Which of the following should the technician do to fix this problem?

- ✓ Replace the rollers
- Replace the toner
- Clean the transfer drum
- Clear the page counter

Feedback

General Feedback

OBJ 5.6: If the rollers become unevenly worn, they will feed the pages in at an angle,

causing the misalignment in the printed output. To fix this issue, the rollers should be replaced. The page, country, toner, and transfer drum will not affect the angle of the printed images on the paper.</p>
For support or reporting issues, include Question ID: 63fe05a63b7322449ddbb7b4 in your ticket. Thank you.

88 1 / 1 point Multiple Choice

A content creator wants to reduce background noise during recordings. Which of the following microphone features would be most beneficial?

- ◯ Stereo recording
- ◯ Wide frequency response
- ✓ ◉ Noise cancellation
- ◯ High sensitivity

Feedback

General Feedback

OBJ 1.1 - Noise cancellation technology helps filter out background noise, improving audio clarity during recordings. High sensitivity increases the microphone's ability to capture sound but can amplify unwanted noise. A wide-frequency response affects the range of sounds captured but does not reduce noise. Stereo recording captures audio in two channels and is unrelated to noise cancellation.
For support or reporting issues, include Question ID: 6797d1feaad21b81318d3e32 in your ticket. Thank you.

89 1 / 1 point Multiple Choice

Which of the following network performance metrics is used to represent the round-trip time it takes for a packet to be sent by a device to a server and then a response received from that destination server?

- ◯ Throughput
- ◯ Bandwidth

- ◯ Jitter
- ✓ ◉ Latency

Feedback

General Feedback

OBJ 5.5: Latency is the measure of time that it takes for data to reach its destination across a network. Usually, we measure network latency as the round-trip time from a workstation to the distant end and back. Bandwidth is the maximum rate of data transfer across a given network. Now, bandwidth is more of a theoretical concept that measures how much data could be transferred from a source to a destination under ideal conditions. Throughput is an actual measure of how much data is successfully transferred from the source to a destination. Therefore, we often measure throughput, instead of bandwidth, to monitor our network performance. Jitter is a network condition that occurs when a time delay in the sending of data packets over a network connection occurs. Jitter is a big problem for any real-time applications you may be supporting on your networks, like video conferences, voice-over IP, and virtual desktop infrastructure clients.
For support or reporting issues, include Question ID: 63fe05773b7322449ddbb57a in your ticket. Thank you.

90 0 / 1 point Multiple Choice

A user reports that the built-in webcam on their laptop is not functioning during video calls. You verify that the webcam is enabled in the BIOS, but the operating system does not detect it. What is the next troubleshooting step?

- ◯ Replace the laptop screen assembly
- ◯ Check the physical connection of the webcam
- ◯ Install third-party software to force the webcam to work
- ✗ ◉ Restart the video-conferencing application

Feedback

General Feedback

OBJ 1.1 - If the webcam is enabled in the BIOS but not detected by the operating system,

it is possible that the physical connection of the webcam is loose or disconnected. Checking this connection is the next logical step. Replacing the screen assembly or using third-party software is unnecessary without further diagnosis. Restarting the application will not help if the operating system cannot detect the hardware.
For support or reporting issues, include Question ID: 6797d49eaad21b81318d3e74 in your ticket. Thank you.